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SRAM macro design architecture

Abstract

A memory device layout that implements SRAM cells with stacked transistors is disclosed. The memory utilizes both topside metal routing and backside metal routing for routing of bitlines between bit cells with stacked transistors and logic cells coupled to the bit cells.

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Background/Summary

PRIORITY CLAIM (1) The present application claims priority to U.S. Provisional App. Ser. No. 63/376,796, entitled “Stacked FET Standard Cell Architecture,” filed Sep. 23, 2022, U.S. Provisional App. Ser. No. 63/376,799, entitled “Stacked SRAM Cell Architecture,” filed Sep. 23, 2022, U.S. Provisional App. Ser. No. 63/376,800, entitled “SRAM Macro Design Architecture,” filed Sep. 23, 2022, U.S. Provisional App. Ser. No. 63/376,802, entitled “Vertical Transistors With Backside Power Delivery,” filed Sep. 23, 2022; the disclosures of each of the above-referenced applications are incorporated by reference herein in their entireties.

BACKGROUND

Technical Field

(1) Embodiments described herein relate to power and signal routing for semiconductor devices. More particularly, embodiments described herein relate to SRAM cells and routing of bitlines to/from SRAM cells through both topside and backside layers.

Description of the Related Art

(2) Standard cells are groups of transistors, passive structures, and interconnect structures that can provide logic functions, storage functions, etc. Current trends in standard cell methodology are towards reducing the size of standard cells while increasing the complexity (e.g., circuit density and number of components or transistors) within standard cells. As standard cell designs become smaller, however, it becomes more difficult to provide access (e.g., connections) to components within the standard cells and within design/manufacturing constraints of standard cells.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features and advantages of the methods and apparatus of the embodiments described in this disclosure will be more fully appreciated by reference to the following detailed description of presently preferred but nonetheless illustrative embodiments in accordance with the embodiments described in this disclosure when taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 depicts a topside plan view representation of a standard cell, according to some embodiments.

(3) FIG. 2 depicts a backside plan view representation of a standard cell, according to some embodiments.

(4) FIG. 3 depicts a cross-sectional representation of a standard cell along the line 3-3 shown in FIG. 1.

(5) FIG. 4 depicts a cross-sectional representation of a standard cell along the line 4-4 shown in FIG. 2.

(6) FIG. 5 depicts a cross-sectional representation of a standard cell along the line 5-5 shown in both FIGS. 1 and 2.

(7) FIG. 6 depicts a cross-sectional representation of a standard cell along the line 6-6 shown in

both FIGS. 1 and 2.

(8) FIG. 7 depicts a topside plan view representation of a cell having alternative vias along the boundary, according to some embodiments.

(9) FIG. 8 is a cross-sectional representation of a cell showing connections to source regions, according to some embodiments.

(10) FIG. 9 is a cross-section representation of a cell showing connections to drain regions, according to some embodiments.

(11) FIG. 10 depicts a cross-sectional representation of a stacked transistor control signal connections in a cell implementing via pillars, according to some embodiments.

(12) FIG. 11 depicts a cross-sectional representation of a stacked transistor control signal connections in a common gate configuration, according to some embodiments.

(13) FIG. 12 depicts a cross-sectional representation of a stacked transistor control signal connections in a cross-coupled gate configuration, according to some embodiments.

(14) FIG. 13 depicts a topside plan view representation of a NAND cell, according to some embodiments.

(15) FIG. 14 depicts a backside plan view representation of a NAND cell, according to some embodiments.

(16) FIG. 15 depicts a cross-sectional representation of a NAND cell along the line A-A' shown in FIGS. 13 and 14.

(17) FIG. 16 depicts a cross-sectional representation of a NAND cell along the line B-B' shown in FIGS. 13 and 14.

(18) FIG. 17 depicts a cross-sectional representation of a NAND cell along the line C-C' shown in FIGS. 13 and 14.

(19) FIG. 18 depicts a cross-sectional representation of a NAND cell along the line D-D' shown in FIGS. 13 and 14.

(20) FIG. 19 depicts a schematic representation of a memory cell.

(21) FIG. 20 depicts a topside plan view representation of a memory cell with stacked transistors, according to some embodiments.

(22) FIG. 21 depicts a backside plan view representation of a memory cell with stacked transistors, according to some embodiments.

(23) FIG. 22 depicts a cross-sectional representation of a memory cell along the line A-A' shown in both FIGS. 20 and 21.

(24) FIG. 23 depicts a cross-sectional representation of a memory cell along the line B-B' shown in both FIGS. 20 and 21.

(25) FIG. 24 depicts a cross-sectional representation of a memory cell along the line C-C' shown in both FIGS. 20 and 21.

(26) FIG. 25 depicts a block diagram representation of a memory device, according to some embodiments.

(27) FIG. 26 depicts a topside plan view representation of a region having dummy cells, according to some embodiments.

(28) FIG. 27 depicts a backside plan view representation of the region having dummy cells, according to some embodiments.

(29) FIG. 28 depicts a cross-sectional representation of the region having dummy cells along the line A-A' shown in both FIGS. 26 and 27.

(30) FIG. 29 depicts a cross-sectional representation of the region having dummy cells along the line B-B' shown in both FIGS. 26 and 27.

(31) FIG. 30 depicts a topside plan view representation of a region having a dummy cell, according to some embodiments.

(32) FIG. 31 depicts a backside plan view representation of the region having a dummy cell, according to some embodiments.

(33) FIG. **32** depicts a cross-sectional representation of the region having a dummy cell along the line A-A' shown in both FIGS. **30** and **31**.

(34) FIG. **33** depicts a cross-sectional representation of the region having a dummy cell along the line B-B' shown in both FIGS. **30** and **31**.

(35) FIG. **34** depicts a schematic representation of a column I/O cell, according to some embodiments.

(36) FIG. **35** depicts a layout of a column I/O cell, according to some embodiments.

(37) FIG. **36** depicts a perspective representation of a contemplated vertical transistor device, according to some embodiments.

(38) FIG. **37** depicts a perspective representation of another contemplated vertical transistor device, according to some embodiments.

(39) FIG. **38** depicts a perspective view representation of an inverter cell construction, according to some embodiments.

(40) FIG. **39** depicts a topside plan view representation of an inverter cell construction, according to some embodiments.

(41) FIG. **40** depicts a backside plan view representation of an inverter cell construction, according to some embodiments.

(42) FIG. **41** depicts a cross-sectional representation of an inverter cell construction, according to some embodiments, along line **41-41** shown in FIG. **39**.

(43) FIG. **42** depicts a cross-sectional representation of an inverter cell construction, according to some embodiments, along line **42-42** shown in FIG. **39**.

(44) FIG. **43** depicts a perspective view representation of a NAND cell construction, according to some embodiments.

(45) FIG. **44** depicts a topside plan view representation of a NAND cell construction, according to some embodiments.

(46) FIG. **45** depicts a backside plan view representation of the NAND cell construction, according to some embodiments.

(47) FIG. **46** depicts a cross-sectional representation of a NAND cell construction, according to some embodiments, along line **46-46** shown in FIG. **44**.

(48) FIG. **47** depicts a cross-sectional representation of a NAND cell construction, according to some embodiments, along line **47-47** shown in FIG. **44**.

(49) FIG. **48** depicts a perspective view representation of a MUX cell construction, according to some embodiments.

(50) FIG. **49** depicts a topside plan view representation of a MUX cell construction, according to some embodiments.

(51) FIG. **50** depicts a backside plan view representation of a MUX cell construction, according to some embodiments.

(52) FIG. **51** depicts a cross-sectional representation of a MUX cell construction, according to some embodiments, along line **51-51** shown in FIG. **49**.

(53) FIG. **52** depicts a cross-sectional representation of a MUX cell construction, according to some embodiments, along line **52-52** shown in FIG. **49**.

(54) FIG. **53** depicts a perspective view representation of a device, according to some embodiments.

(55) FIG. **54** depicts a cross-sectional representation of a device, according to some embodiments, along line **54-54** shown in FIG. **53**.

(56) FIG. **55** is a block diagram of one embodiment of an example system.

(57) Although the embodiments disclosed herein are susceptible to various modifications and alternative forms, specific embodiments are shown by way of example in the drawings and are described herein in detail. It should be understood, however, that drawings and detailed description thereto are not intended to limit the scope of the claims to the particular forms disclosed. On the

contrary, this application is intended to cover all modifications, equivalents and alternatives falling within the spirit and scope of the disclosure of the present application as defined by the appended claims.

DETAILED DESCRIPTION OF EMBODIMENTS

(58) As used herein, the term “standard cell” refers to a group of transistor structures, passive structures, and interconnect structures formed on a substrate to provide logic or storage functions that are standard for a variety of implementations. For example, an individual standard cell may be one cell in a library of multiple cells from which various suitable cells may be selected to implement a specific cell design. Integrated circuit cells may also include custom circuit design cells that are individually designed for a particular implementation. Embodiments of circuit design cells described herein may be implemented in various implementations of logic integrated circuits or memory integrated circuits.

(59) Many current designs of cells provide connections and routing for power or signals to transistors or other structures in areas above the transistors. For example, the connections and routing for power or signals may be provided in topside layers of the device. As used herein, the term “topside” refers to areas in a device that are vertically above an active layer of the device (e.g., above a transistor region of the device when viewed in a typical cross-sectional view). For example, topside may refer to components such as contacts or layers that are above a transistor region in a vertical dimension, as depicted in the figures and described herein. In some instances, the term “frontside” may be used interchangeably with the term “topside”.

(60) Some recent developments for designs of standard cells move connections and routing for power connections to metal layers below the transistors. For example, the connections and routing for power may be provided in the backside layers of the device. As used herein, the term “backside” refers to areas in a device that are vertically below an active layer of the device (e.g., below a transistor region of the device when viewed in a typical cross-sectional view). For example, backside may refer to components such as contacts or layers that are below a transistor region in a vertical dimension, as depicted in the figures and described herein. It is noted that as used herein, backside elements located below an active layer may be situated above, within, or below a silicon substrate on which the active layer is manufactured. That is, as used herein, “backside” is relative to the active layer, rather than the silicon substrate.

(61) The present disclosure is directed to various implementations of stacked transistors or vertical transistors in integrated circuit cells (e.g., standard cells) that utilize connections to both topside metal layers and backside metal layers. The present inventors have recognized that topside and backside layers can be utilized in specific ways to provide technical and space saving advantages for cell layouts implementing stacked transistors or vertical transistors. The disclosed embodiments implement topside and backside metal layers to provide advantageous cell layouts and routing (e.g., paths) for control signals or power signals within the cell layouts.

(62) Stacked Transistor Standard Cell Designs

(63) Stacked transistors (e.g., where two transistor active regions are stacked vertically above a substrate) may provide various technical and space saving advantages due to the proximity of devices in the transistors. The implementation of stacked transistors in a standard cell, however, is challenging due to design and manufacturing constraints associated with standard cell construction. For example, in standard cells that utilize only topside routing, there are typically not enough paths for routing to both transistors without expanding the size of the standard cell. Standard cells that are limited to topside routing for control signals and backside routing for power signals may also lack the necessary routing and connection availability for two stacked transistors without changes to the size of the standard cell.

(64) The present disclosure contemplates various techniques that implement routing in both topside and backside metal layers that allow two stacked transistors to be placed in a standard cell. Various embodiments of standard cell constructions are disclosed that provide basic building blocks for

many different types of devices from simple devices (e.g., inverters and NAND devices) to more complex devices (e.g., complex FETs). The disclosed embodiments provide a compact standard cell construction that allows for the implementation of stacked transistors in various circuit logic schemes.

(65) Certain embodiments disclosed herein have four broad elements: 1) a first metal layer located above a transistor region of an integrated circuit cell structure (e.g., a topside metal layer); 2) a second metal layer located below the transistor region (e.g., a backside metal layer), 3) a pair of vertically stacked transistors in the transistor region, and 4) various possible connection paths for both control signals and power signals between either the first or second metal layers and the first and second transistors. In certain embodiments, the pair of vertically stacked transistors includes heterogeneous transistors (e.g., complementary transistor types such as PMOS and NMOS transistors). In some embodiments, the pair of vertically stacked transistors includes homogeneous transistors (e.g., the transistors are of the same type).

(66) In various embodiments, the control signal and power signal connections are made to implement logic associated with specific integrated circuit devices having multiple transistors for the standard cell constructions described herein. For instance, examples of an inverter device or a NAND gate device that may be implemented based on the standard cell construction are described below. Embodiments of various possible connections for control signals and voltage signals to the stacked transistors within the standard cell construction are also described. A person with knowledge in the art would understand that combinations of these various possible connections may be implemented to generate many different desired circuits based on the stacked transistor structure within the standard cell construction.

(67) In short, the present inventors have recognized that providing various routing paths within a standard cell construction allows the implementation of connections to both topside and backside metal layers for control signals and power signals from stacked transistors positioned within the standard cell construction. The routing paths described herein enable standard cell constructions to be utilized in generating a variety of both simple and complex integrated circuit logic devices based on the stacked transistors within the cells. Additionally, the standard cell construction with stacked transistors described herein provides a scalable template that can be implemented in devices having multiple integrated circuit cells. The standard cell construction with stacked transistors within the present disclosure further enables constructions of cells that can be implemented within current manufacturing constraints and without changes to the size or parameters of current standard cells. As used herein, the term “routing” refers to any combination of metal vias, metal wires, metal traces, etc. that provide a path/route between two structures. Additional embodiments may be contemplated where the metal in “routing” is replaced with an alternative conductive material. For instance, the metal in “routing” may be replaced with a superconductor material, a semiconductor material, or a non-metal conductor.

(68) FIGS. 1-6 depict representations of a standard cell with stacked transistors and both topside and backside layer connections, according to some embodiments. FIG. 1 depicts a topside plan view representation of standard cell 100, according to some embodiments. FIG. 2 depicts a backside plan view representation of standard cell 100, according to some embodiments. FIG. 3 depicts a cross-sectional representation of standard cell 100 along the line 3-3 shown in FIG. 1. FIG. 4 depicts a cross-sectional representation of standard cell 100 along the line 4-4 shown in FIG. 2. FIG. 5 depicts a cross-sectional representation of standard cell 100 along the line 5-5 shown in both FIGS. 1 and 2. FIG. 6 depicts a cross-sectional representation of standard cell 100 along the line 6-6 shown in both FIGS. 1 and 2.

(69) For simplicity in the drawings, only components relevant to the disclosure are shown in the representations of a cell disclosed herein. A person with knowledge in the art would understand that additional components may be present in any of the cells depicted herein. For instance, in FIG. 1, various connections (such as vias or contacts described herein) may be visible in some depictions.

Additionally, some transparency of material is provided to enable visibility of underlying components in topside and backside plan views for better understanding of the disclosed embodiments. For instance, in FIGS. 1 and 2, substrate **102** has some transparency to provide visibility of gates and active regions of the underlying transistors and topside metal layers **112** and backside metal layers **120** have some transparency to provide visibility of contact **114** and backside via **122**, respectively.

(70) In various embodiments, as shown in FIGS. 1-6, standard cell **100** includes substrate **102**. In certain embodiments, substrate **102** is a silicon substrate though other semiconductor substrates may also be contemplated. Substrate **102** may include additional components or features for implementation in cell **100**. For instance, substrate **102** may include one or more insulating layers (e.g., oxide layers), diffusion (e.g., oxide diffusion) regions, or doped regions for implementation in cell **100**. For simplicity in the drawings, substrate **102** is depicted as a material filling the volume of standard cell **100**.

(71) In various embodiments, first active region **104** and second active region **106** are formed in substrate **102**. In certain embodiments, active region **104** is positioned vertically above active region **106** in cell **100**. For example, active region **104** may be positioned in an upper portion of cell **100** with active region **106** in a lower portion of the cell. It should be noted that active region **104** and active region **106** may not necessarily be positioned directly above/beneath each other. For instance, some portions of active region **104** or active region **106** may be outside the boundaries of the other active region. Thus, when describing that active region **104** is above active region **106**, it may be referencing that at least some portion of active region **104** is above at least some portion of active region **106**, or vice versa.

(72) In certain embodiments, active region **104** is an active region of a first transistor in cell **100** and active region **106** is an active region of a second transistor in the cell. For example, in one contemplated embodiment, active region **104** is the active region of an NMOS transistor with one or more NMOS gates and active region **106** is the active region of a PMOS transistor with one or more PMOS gates. Alternatively, active region **104** may be the active region of a PMOS transistor while active region **106** is the active region of an NMOS transistor. While transistor stacks with these complementary transistor types (e.g., heterogeneous transistors) are described with respect to FIGS. 1-6, it should be understood that additional embodiments may be contemplated where both transistors are of the same type (e.g., homogeneous transistors). Additionally, while the embodiments disclosed describe silicon-based transistors such as NMOS and PMOS transistors, other types of semiconductor-based transistors may be contemplated without deviating from the scope of the present disclosure. Yet further, any types of transistor structures may be contemplated. For instance, the transistors formed may include transistors such as, but not limited to, FinFETs, nanosheet FETs (NSHs), or GAAFETs (“gate-all-around” FETs).

(73) In various embodiments, transistors with active region **104** include upper gates **108** and source/drain regions **124** while transistors with active region **106** include lower gates **116** and source/drain regions **124**. The transistors may be positioned between isolation gates **110**, as shown in FIGS. 1-4. Cell **100** may also include contacts **126** that provide connections for source/drain regions **124**. Cell **100**, as depicted in FIGS. 1-6, includes two upper gates **108A-B**, two lower gates **116A-B**, six source/drain regions **124A-F** (with three source/drain regions **124A-C** in the upper transistor and three source/drain regions **124D-F** in the lower transistor), and six contacts **126A-F** positioned between isolation gates **110** in the cell **100**. The embodiment of cell **100** depicted in FIGS. 1-6 is, however, one example of a possible construction of transistor components within the cell. For instance, it should be understood that cell **100** may include any number of transistor components that fit within the boundaries of the cell according to design and manufacturing constraints of the cell (e.g., the design and manufacturing constraints of a standard cell) without deviating from the scope of the present disclosure.

(74) Upper gates **108** and lower gates **116**, may be, for example, poly lines (e.g., polysilicon layers)

or high-k/metal gates. In certain embodiments, upper gates **108** and lower gates **116** include gate spacers (not shown for simplicity in the drawing). For instance, gate spacers may be positioned between gates **108/116** and source/drain regions **124**. In various embodiments, the gate spacers are formed as parts of gates **108/116** (e.g., the gates and spacers are formed in the same process flow). Source/drain regions **124** may be, for example, epitaxial layers grown on fins or nanosheet stacks or any 2D (two-dimensional) channel materials. Various embodiments may also be contemplated where source/drain regions **124** are at least partially positioned in substrate **102**.

(75) In addition to the above-described transistor components, various connections to gates and/or source/drain regions may be made within cell **100**. The connections may include, for example, contacts or vias that provide connection between transistor components in cell **100** and top-side metal layers **112** or backside metal layers **120**. Contact **114** (shown in FIGS. **1**, **3**, and **5**) is an example of a gate contact that provides connection between upper gate **108A** and top-side metal layer **112B**. Backside via **122** (shown in FIGS. **2**, **4**, and **6**) is an example of a via that provides connection between source/drain region **124F** and backside metal layer **120B**.

(76) In various embodiments, contacts **114** are top-side vias that provide connection to top-side metal layers **112** from various portions of the transistors (e.g., source/drain regions, gates, etc.) while backside vias **122** provide connection to backside metal layers from various portions of the transistors. Contacts **114** and backside vias **122** are provided as non-limiting examples of connections possible in cell **100**. For instance, any number or combination of contacts or vias (along with any horizontal routing) may be implemented in cell **100** to provide connections between various top-side metal layers and backside metal layers and transistor components (e.g., upper gates **108**, lower gates **116**, and source/drain regions **124**) within the cell. Connections may also be made depending on whether control signals or power signals are intended for components within the transistors of cell **100**. For example, in the embodiment depicted in FIGS. **1-6**, contact **114** may provide a control signal connection to upper gate **108A** while backside via **122** provides a power signal connection to source/drain region **124F**. The number, types, and positioning of contacts and vias may be determined based on the desired device being constructed utilizing the component structure inside cell **100**. For instance, an inverter device has different connections from a NAND device (shown as example in FIGS. **13-18**).

(77) As shown in FIGS. **1-6**, cell **100** includes four top-side metal tracks (top-side metal layers **112A-D**) and three backside metal tracks (backside metal layers **120A-C**). It should be understood that cell **100** may include any number of top-side metal tracks and backside metal tracks as allowed according to design and manufacturing constraints of the cell. For instance, the number of metal tracks may be higher or lower depending on design and manufacturing constraints for the cell such as height, pitch, width, etc. Additionally, top-side metal tracks and backside metal tracks may not be aligned with each other. For instance, as clearly depicted in FIGS. **5-6**, top-side metal layers **112** are not aligned with backside metal layers **120**. Embodiments with aligned top-side and backside metal layers may, however, be contemplated.

(78) In various embodiments, as shown in FIG. **4**, backside metal layers **120** are formed at or near a bottom surface of substrate **102**. In certain embodiments, backside metal layers **120** include one or more backside layers of an active layer in cell **100** (e.g., a backside metal layer is vertically below active region **106**). In some embodiments, backside metal layers **120** include one or more buried layers in substrate **102** (e.g., the metal layers are buried or embedded underneath the bottom surface of the substrate). In some embodiments, backside metal layers **120** are buried beneath a carrier substrate layer (e.g., a silicon carrier substrate). Additional embodiments may be contemplated where backside metal layers **120** are not located in substrate **102**.

(79) Metal tracks also may be selected for use as control signal tracks (e.g., control signal rails) or power signal tracks (e.g., power signal rails) based on the desired device usage of cell **100**. Control signal tracks may provide input or output signal connections to transistor components in cell **100** while power signal tracks may provide power routing to/from Vdd (e.g., the supply voltage) and

Vss (e.g., ground) as well as other contemplated power supply connections. The selection of a track for use as a control signal track or power signal track may also determine connections made to the metal track. For instance, in the depicted embodiment of cell **100**, topside metal layer **112B** is a control signal track connected to upper gate **108A** by contact **114** while backside metal layer **120B** is a power signal track connected to source/drain region **124F** by backside via **122**. It should be noted that any of backside metal layers **120A-C** may be used as a power signal track in various contemplated embodiments. The remaining topside metal tracks (e.g., topside metal layers **112A**, **112C**, **112D**) may be used for either additional topside control signal tracks or topside power signal tracks. Similarly, the remaining backside metal tracks (e.g., backside metal layers **120A** and **120C**) may be used for either additional backside power signal tracks or backside control signal tracks. For instance, in one contemplated embodiment, one of topside metal layers **112A**, **112C**, **112D** may be a topside power signal track that has a contact to a source/drain region in the upper portion of cell **100** (e.g., one of source/drain region **124A-C**) to provide a power signal connection to the upper transistor with active region **104**. Additionally, one of backside metal layers **120A** and **120C** may be a backside control signal track that has a contact to lower gate **116B** (e.g., the lower gate in the split gate configuration) in the lower portion of cell **100** to provide a control signal connection to the lower transistor with active region **106**.

(80) In various contemplated embodiments, one or more topside metal tracks (e.g. topside metal layers **112A-D**) may be merged to when the metal tracks are being used as power rails (e.g., power signal tracks). For instance, in some embodiments, when a metal track along the cell edge (e.g., topside metal layer **112A** or topside metal layer **112D**) is used as a power rail, the metal track may be merged with the metal cell track from the neighboring cell. For example, as shown in FIG. **1**, topside metal layer **112D'** may be in the neighboring cell in the cell height direction (above cell **100** in the depiction of FIG. **1**). Power rail merge layer **109** (dashed line box) may then be implemented to merge topside metal layer **112A** in cell **100** with topside metal layer **112D'** in the above neighboring cell. In various embodiments, power rail merge layer **109** is a metal layer connecting topside metal layer **112A** and topside metal layer **112D'**. In some contemplated embodiments, topside metal layer **112A**, topside metal layer **112D'**, and power rail merge layer **109** may be a single metal track formed in cell **100** and the neighboring cell.

(81) In other contemplated embodiments for merging of metal tracks, metal tracks at or near a center of a cell that are used as power rails may be merged. For instance, as shown in FIG. **1**, power rail merge layer **111** (dashed line box) may be implemented to merge topside metal layer **112B** and topside metal layer **112C** when these layers are used as power rails. Power rail merge layer **111** may be, for instance, metal connecting topside metal layer **112B** and topside metal layer **112C**. In some instances, topside metal layer **112B**, topside metal layer **112C**, and power rail merge layer **111** may be a single metal track in cell **100**.

(82) It should be noted that in the embodiment of cell **100** depicted in FIGS. **1-6**, additional contacts for control signals and power signals are not shown for simplicity in the drawings with the understanding that various connections could be made between the depicted metal tracks and the transistor components within the cell. For instance, any metal track (e.g., any one of topside metal layers **112A**, **112C**, **112D** or any one of backside metal layers **120A** and **120C**) can be connected to any gate (e.g., upper gate **108** or lower gate **116**) or source/drain region **124** within cell **100** as needed for a desired device structure. Thus, the depicted structure of cell **100** in FIGS. **1-6** with components (e.g., gates **108/116** and source/drain regions **124**) and metal tracks (e.g., topside metal layers **112** and backside metal layers **120**) provides a basic building block structure from which different variations of connections between the components and the metal tracks can be made to generate many different types of devices.

(83) As shown in FIGS. **1-6**, upper gates **108** and lower gates **116** may be formed in cell **100** with different relationships between the gates in the stacked transistor structure. Two possible embodiments for gate relationships are depicted in FIGS. **3** and **4**. In a first contemplated

embodiment, upper gate **108A** is merged with lower gate **116A**, creating connectivity between the gates. The connectivity created by merging of the gates allows a single control signal to be provided to both gates. For example, as shown in FIGS. **3** and **4**, a control signal provided to upper gate **108A** from topside metal layer **112B** through contact **114** may be passed through and used as the control signal for lower gate **116A** because of the connectivity between the gates. It should be noted that the merged gate configuration is also shown in the cross-section representation of FIG. **5**. A second contemplated embodiment includes upper gate **108B** being separated (e.g., split) from lower gate **116B**. Separating upper gate **108B** and lower gate **116B** keeps the gates electrically disconnected. Thus, upper gate **108B** and lower gate **116B** would need separate control signals since a control signal cannot be passed between the gates due to the separation/split between the gates.

(84) While not shown in FIGS. **1-6**, source/drain regions **124** may also be merged between the upper and lower transistors in various embodiments. Examples of merged source/drain regions are shown in FIGS. **13-16**, described below. Merging of source/drain regions **124** may allow for a single power signal to connect power to both the upper and lower transistors or a single control signal (e.g., a single output signal from a drain) to be connected to both the upper and lower transistors. Similar to splitting of gates, separate connections need to be made to the split source/drain regions when source/drain regions **124** are split (as shown in FIGS. **1-6**).

(85) In certain embodiments, power routing for the transistors in cell **100** (e.g., the upper transistor and the lower transistor) is divided between the topside metal layers **112** and the backside metal layers **120**. For instance, as shown in FIGS. **2, 4, 6**, backside metal layer **120B** is connected to source/drain region **124F** through backside via **122**. Thus, in certain embodiments, power to lower gate **116B** in the lower transistor (e.g., the transistor with active region **206**) is routed from backside metal layer **120B**. To divide power, power to upper gate **108B** may be routed from one of topside metal layers not used for control signals (such as, for example, topside metal layer **112C**). Accordingly, upper gate **108B** and lower gate **116B** would have separate power rails providing power to the gates/transistors. Dividing the power routing between the topside metal layers **112** and the backside metal layers **120** may provide more efficient power routing in cell **100**.

(86) Additional embodiments may be contemplated where power routing is provided from one or the other of the topside metal layers **112** and the backside metal layers **120** but not both layers. For instance, as discussed above, the source/drain regions could be merged and a single power signal from either the topside or backside could be provided to the merged regions. In another contemplated embodiment, power may be brought into cell **100** through a backside metal layer (such as backside metal layer **120A**) and a via could route power from the backside metal layer to one of the topside metal layers, which then connects to a source/drain region in the upper transistor. The via could be positioned in an open channel in cell **100** between topside metal layers and backside metal layers. One example of a via is shown in FIG. **6** where via **600** (dashed lines) is optionally routed between topside metal layer **112A** and backside metal layer **120A** in the space (e.g., channel) outside source/drain regions **124C/124F**. Topside metal layer **112A** may then connect (directly or through another topside metal layer) to source/drain region **124C** to provide power in the upper transistor. Note that via **600** as shown in FIG. **6** can also be used to route signals between topside and backside metal layers.

(87) Yet another alternative for routing power from backside layers to the upper could include extending source/drain region **124C** to intersect and connect with via **600**, as shown by the dotted lines extending from source/drain region **124C** in FIG. **6**. The extension would create a direct connection between via **600** and source/drain region **124C**. Other examples of extensions (both gates and source/drain regions) are provided below in reference to FIGS. **7-12**, which describe the implementation of via pillars in cells (e.g., on the perimeter of the cells).

(88) As discussed above, the embodiment of cell **100** depicted in FIGS. **1-6** provides a basis for a compact standard cell structure that implements two vertically stacked transistors. The transistors

can be heterogeneous (e.g., two different types of transistors) or homogeneous (e.g., the same type of transistor). Cell **100** implements availability for connections from either topside metal layers and backside metal layers to any of the various transistor components including the gates and the source/drain regions. The adaptability in connections for both control signal and power for the transistor components allows the basic structure of cell **100** to be adapted to a wide range of logic schemes to implement different integrated circuit devices using stacked transistors. The compactness of cell **100** allows a cell with stacked transistors to be implemented within current standard cell design and manufacturing constraints.

(89) In some instances, placing cell **100** next to neighboring cells may cause manufacturing issues due to routing tracks along the boundaries of the cells. With the routing tracks along the boundaries, vias from the routing tracks may create spacing issues between the neighboring cells. One possible solution to this boundary problem is to replace the routing tracks along the edges with via tracks where the vias alternate between being used by the current cell and the neighboring cell.

(90) FIG. 7 depicts a topside plan view representation of a cell having alternative vias along the boundary, according to some embodiments. In cell **700**, three topside metal layers **112A**, **112B**, **112C** are positioned between via track **705A** and via track **705B**. Via track **705A** and via track **705B** are placed along the boundary (e.g., edge) of cell **700**. Via tracks **705A** and via track **705B** include via pillars **710**. In certain embodiments, via pillars **710A** and via pillars **710B** with via pillars **710A** belonging to cell **700** while via pillars **710B** belong to neighboring cells. In various embodiments, via pillars **710** placed on the perimeter of cells and are utilized to provide routing into the interior of cell **700** for either control signals or power signals. It should be noted that neighboring cells may have mirrored setups to cell **700** to accommodate the alternating via pillar arrangement.

(91) Various embodiments for connecting to via pillars **710A** within cell **700**, shown in FIG. 7, may be contemplated. Embodiments may include routing connections to gates or source/drain regions within cell **700**. FIGS. 8 and 9 depict cross-sectional representations of source and drain connections in a cell implementing via pillars, according to some embodiments. FIG. 8 is a cross-sectional representation of cell **800** showing connections to source regions **124A**, **124B** while FIG. 9 is a cross-section representation of cell **800** showing connections to drain regions **124C**, **124D**.

(92) In various embodiments, as shown in FIG. 8, source regions **124A**, **124B** are connected directly to topside metal layer **112B** and backside metal layer **120A**, respectively by contacts **114**. Topside metal layer **112B** and backside metal layer **120A** may then route to power supply (e.g., Vdd) or ground (e.g., Vss) for power connections to source regions **124A**, **124B**. For drain regions **124C**, **124D**, as shown in FIG. 9, contacts **114** are routed horizontally to via pillar **710A**. Via pillar **710A** may then route to topside metal layer **112** or backside metal layer **120** for signal connection. In some embodiments, contacts **114** for drain regions **124C**, **124D** may be replaced by extending drain regions **124C**, **124D** horizontally to connect with via pillar **710A**.

(93) FIG. 10 depicts a cross-sectional representation of a stacked transistor control signal connections in a cell implementing via pillars, according to some embodiments. In FIG. 10, cell **1000** includes source/drain regions **124A**, **124B** connected to topside metal layer **112C** and backside metal layer **120B**, respectively, by contacts **114**. Topside metal layer **112C** and backside metal layer **120B** may be routing for control signals provided to source/drain regions **124A**, **124B**.

(94) Via pillars **710A** may also provide routing for control signals from gates in various embodiments of stacked transistors. The gates may, for example, be coupled to as common gates or split gates (e.g., as used in transmission gates). FIG. 11 depicts a cross-sectional representation of a stacked transistor control signal connections in a common gate configuration, according to some embodiments. In cell **1100**, upper gate **108A** and lower gate **116A** are extended to intersect and connect to via pillar **710A** for a common connection to the gates. FIG. 12 depicts a cross-sectional representation of a stacked transistor control signal connections in a split-gate configuration, according to some embodiments. In cell **1200**, upper gate **108A** is connected to topside metal layer **112C** by contact **114** and lower gate **116A** is extended to intersect and connect to via pillar **710A**.

Topside metal layer **112C** and via pillar **710A** may then be cross-coupled to cross-couple upper gate **108A** and lower gate **116A**.

(95) Example NAND Cell with Stacked Transistors

(96) FIGS. **13-18** depict representations of an example NAND cell, according to some embodiments. FIG. **13** depicts a topside plan view representation of NAND cell **1300**, according to some embodiments. FIG. **14** depicts a backside plan view representation of NAND cell **1300**, according to some embodiments. FIG. **15** depicts a cross-sectional representation of NAND cell **1300** along the line A-A' shown in FIGS. **13** and **14**. FIG. **16** depicts a cross-sectional representation of NAND cell **1300** along the line B-B' shown in FIGS. **13** and **14**. FIG. **17** depicts a cross-sectional representation of NAND cell **1300** along the line C-C' shown in FIGS. **13** and **14**. FIG. **18** depicts a cross-sectional representation of NAND cell **1300** along the line D-D' shown in FIGS. **13** and **14**.

(97) In certain embodiments, FIG. **13** depicts (from the topside) structures associated with active region **104** of the upper transistor (e.g., the NMOS active region) and FIG. **14** depicts (from the backside) structures associated with active region **106** of the lower transistor (e.g., the PMOS active region) in the vertically stacked transistors. In the illustrated embodiment, topside metal layer **112D** is a ground rail (e.g., VSS rail) while backside metal layer **120B** is a power supply rail (e.g., VDD rail). The remaining topside metal layers (e.g., topside metal layers **112A**, **112B**, **112C**) and backside metal layers (e.g., backside metal layers **120A**, **120C**) may be used for signal routing and/or internal routing inside NAND cell **1300**, as described below.

(98) In the illustrated embodiment, as shown in FIG. **13**, NAND cell **1300** includes upper gate **108A** and upper gate **108B**, which are active NMOS gates positioned inside the cell. Isolation gates **110A**, **B** (e.g., dummy gates) are positioned at opposing ends of cell **1300** in the gate pitch direction while upper contact **126A** is positioned between isolation gate **110A** and upper gate **108A**, upper contact **126B** is positioned between upper gate **108A** and upper gate **108B**, and upper contact **126C** is positioned between upper gate **108B** and isolation gate **110B** also in the gate pitch direction. Further in the illustrated embodiment, as shown in FIG. **14**, NAND cell **1300** includes lower gate **116A** and lower gate **116B**, which are active PMOS gates positioned inside the cell. Lower contact **126D** is positioned between isolation gate **110A** and lower gate **116A**, lower contact **126E** is positioned between lower gate **116A** and lower gate **116B**, and lower contact **126F** is positioned between lower gate **116B** and isolation gate **110B** in the gate pitch direction.

(99) In various embodiments, as shown in FIGS. **13** and **15**, contact **114A** provides connection between contact **126A** and topside metal layer **112D** (e.g., the ground rail). Contact **126A** is also connected to source/drain region **124A** of upper gate **108A**. Note that FIGS. **15-18** depict six source/drain regions **124A-F** in NAND cell **1300** where source/drain regions **124A-C** are upper source/drain regions and source/drain regions **124D-F** are lower source/drain regions. In NAND cell **1300**, upper source/drain regions **124A-C** are separated from lower source/drain regions **124D-F**. Additionally, upper source/drain regions **124A-C** and upper gates **108A**, **108B** include upper channels **1304** while lower source/drain regions **124D-F** and lower gates **116A**, **116B** include lower channels **1306**.

(100) Contact **114B**, shown in FIGS. **13** and **16** provides connection between upper gate **108A** and topside metal layer **112B**, which may be a route for an input signal to the device of NAND cell **1300**. NAND cell **1300** further includes contact **114D** (shown in FIGS. **13** and **18**) that connects contact **126C** to topside metal layer **112A**, which may be a route for an output signal from the device of NAND cell **1300**.

(101) In the contemplated embodiment of NAND cell **1300**, the upper gates and lower gates are merged. For instance, as shown in FIG. **16**, upper gate **108A** is merged to lower gate **116A**. Thus, contact **114B** provides a connection between a merged upper gate **108A** and lower gate **116A** and topside metal layer **112B** (e.g., the input signal route). Upper gate **108B** and lower gate **116B** may be similarly merged and connected to contact **114C** (shown in FIG. **13**), which then provides

connection to topside metal layer **112C**, which may be a second route for an input signal to the device of NAND cell **1300**.

(102) In various embodiments, NAND cell **1300** includes backside via **122A** and backside via **122C**, shown in FIGS. **14**, **15**, **17**, and **18**. Backside via **122A** provides a connection between lower contact **126D** and backside metal layer **120C**, as shown in FIGS. **14** and **15**. Backside metal layer **120C** is also connected to lower contact **126F** by backside via **122C**, as shown in FIGS. **14** and **18**. Accordingly, backside metal layer **120C** provides an internal (to the cell) route path between lower contact **126D** (which is connected to lower source/drain region **124D** of lower gate **116A**) and lower contact **126F** (which is connected to lower source/drain region **124F** of lower gate **116B**).

(103) In certain embodiments, NAND cell **1300** includes via **1302A** and via **1302B**. Via **1302A** and via **1302B** are top-to-back vias that connect upper contact **126C** to lower contact **126F**, as shown in FIGS. **13**, **14**, and **18**. Accordingly, via **1302A** and via **1302B** provide connection between upper source/drain region **124C** of upper gate **108B** and lower source/drain region **124F** of lower gate **116B**. Note that in the depictions of FIGS. **13** and **14**, via **1302A** and via **1302B** may be partially hidden from view by contact **114D** and backside via **122C**, respectively.

(104) NAND cell **1300** further includes backside via **122B**, shown in FIGS. **14** and **17**. Backside via **122B** provides a power supply connection (from backside metal layer **120B**, which is the power supply rail) to lower source/drain region **124E**. Lower source/drain region **124E** is shared by lower gate **116A** and lower gate **116B** in lower active region **106**. Accordingly, power supply is provided to all the active gates through backside via **122B** since lower gate **116A** and lower gate **116B** are respectively merged to upper gate **108A** and upper gate **108B**.

(105) Complementary FET SRAM Cells

(106) In various embodiments, stacked transistors, such as those described herein, may be implemented in memory cells such as SRAM cells. FIG. **19** depicts a schematic representation of a memory cell. Cell **1700** is, by example, a 6T SRAM memory cell. Cell **1700** includes two inverters **1710A**, **1710B** that are cross-coupled with the inputs being fed into the outputs at node **1712** and node **1714**. Node **1712** is coupled to the output of pass gate **1720** and node **1714** is coupled to output of pass gate **1722**. Pass gates **1720**, **1722** may sometimes be referred to as “access gates” or “transmission gates”. Wordline **1730** is coupled to pass gate **1720** and pass gate **1722** to provide control signals to the pass gates. Bitline **1740** is coupled to pass gate **1720** to read/write data from the pass gate. Bitline **1742**, which is complementary to bitline **1740**, is coupled to pass gate **1722** to read/write data from the pass gate.

(107) With inverter **1710A**, inverter **1710B**, pass gate **1720**, and pass gate **1722**, cell **1700** includes six transistors—two in each of the inverters and one in each of the pass gates. In certain embodiments, the inverters **1710** include two complementary transistors—for example, each inverter includes an NMOS transistor and a PMOS transistor. Pass gates **1720**, **1722** may be transistors of the same type. In one embodiment, both pass gates are NMOS transistors. Thus, in various embodiments, cell **1700** includes four NMOS transistors and two PMOS transistors.

(108) The present disclosure contemplates various techniques that implement stacked transistors in a memory cell. For instance, stacked transistors, such as those described above, may be implemented in the 6T SRAM memory cell shown in FIG. **19**. Utilizing the disclosed embodiments of stacked transistors in a memory cell provides capability for minimizing spacing along with multiple transistors in the memory cell. Accordingly, the disclosed embodiments of memory cells include multiple transistors in a small scale factor.

(109) Certain embodiments disclosed herein have five broad elements: 1) a first transistor region with first and second active regions in parallel; 2) a second transistor region with third and fourth active regions in parallel where the second transistor region is positioned vertically below the first transistor region, 3) a first inverter formed by a transistor in the first active region and a transistor in the third active region, 4) a second inverter formed by a transistor in the second active region and a transistor in the fourth active region, and 5) a cross-coupling between the first inverter and

the second inverter. In certain embodiments, the source/drain regions in the inverters are merged. For instance, the source/drain regions on opposite sides of the gates of the two transistors in the first inverter may be merged. In certain embodiments, the cross-coupling is achieved by coupling a horizontally extended portion of the gate for the transistor in the third active region with a source/drain region of the transistor in the fourth active region and coupling a horizontally extended portion of the gate for the transistor in the fourth active region with a source/drain region of the transistor in the third active region.

(110) In various embodiments, the horizontally extended portions of the gates for cross-coupling are portions that extend towards and possibly into the other active region (e.g., the gate for the transistor in the third active region has a portion extending into the fourth active region). Extending the gates as described herein allows the cross-couple connections to be made in areas of the cell vertically below the active regions of the memory cell. The cross-couplings may also be positioned vertically above any backside layer routing. This area is available for cross-coupling by removing material in inactive portions of the third and fourth active regions.

(111) In short, the present inventors have recognized that stacked transistors may be implemented in a memory cell along with the removal of material for inactive portions of active regions to enable cross-coupling of inverters formed by the transistors in the memory cell. The cross-coupling of the inverters in the areas intended for inactive portions provides a memory cell construction that maintains current design philosophies while also reducing the cell height versus typical memory cells. The cell height may be reduced as the utilization of the cross-coupling in an area below the active regions allows the active regions to be vertically positioned closer together. Minimizing the vertical spacing between the active regions accordingly allows reduction in the overall height of the memory cell.

(112) FIG. 20 depicts a topside plan view representation of memory cell **1800** with stacked transistors, according to some embodiments. FIG. 21 depicts a backside plan view representation of memory cell **1800** with stacked transistors, according to some embodiments. FIG. 22 depicts a cross-sectional representation of memory cell **1800** along the line A-A' shown in both FIGS. 20 and 21. FIG. 23 depicts a cross-sectional representation of memory cell **1800** along the line B-B' shown in both FIGS. 20 and 21. FIG. 24 depicts a cross-sectional representation of memory cell **1800** along the line C-C' shown in both FIGS. 20 and 21.

(113) For simplicity in the drawings, only components relevant to the disclosure are shown in the representations of a cell disclosed herein. A person with knowledge in the art would understand that additional components may be present in any of the cells depicted herein. For instance, in FIGS. 20 and 21, various connections (such as vias or contacts described herein) may be visible in some depictions. Additionally, some transparency of material is provided to enable visibility of underlying components in topside and backside plan views for better understanding of the disclosed embodiments. For instance, in FIGS. 20 and 21, gates (e.g., poly lines) and source/drain regions have some transparency to provide visibility of vias/contacts, and active regions in the underlying areas of the transistors and, in FIGS. 20 and 21, the topside and backside metal layers have transparency to provide visibility of the transistors that would be hidden in the plan views. Depths of the various components may be seen more clearly in the cross-sectional representations of FIGS. 22-24. It should be noted that the topside metal layers are not depicted in the cross-sectional representations of FIGS. 22-24 for further simplicity in the drawings.

(114) In the illustrated embodiments, cell **1800** includes two upper active regions **1810**, **1820** (shown in FIG. 20) and two lower active regions **1830**, **1840** (shown in FIG. 21). In certain embodiments, upper active regions **1810**, **1820** are active regions for NMOS transistors and lower active regions **1830**, **1840** are active regions for PMOS transistors. Lower active regions **1830**, **1840** may include inactive portions **1833**, **1842** (shown by the angled fill pattern in the active regions of FIG. 21). Inactive portions **1833**, **1842** may be formed by not having diffusion material in the portions (e.g., through either removing diffusion material or not having diffusion material

deposited in the portions) or disconnecting diffusion material from active portions of the lower active regions **1830**, **1840** (e.g., by an isolation structure or mechanism).

(115) Upper active region **1810** is separated from upper active region **1820** by a diffusion to diffusion spacing distance **1815**. Similarly, lower active region **1830** is separated from lower active region **1840** by a diffusion to diffusion spacing distance **1835**. In some embodiments, distance **1815** and distance **1835** are substantially the same distances.

(116) In the illustrated embodiments, upper active region **1810** includes upper gate **1812** between source/drain region **1814A** and source/drain region **1814B** and upper gate **1816** between source/drain region **1814B** and source/drain region **1814C**. Upper active region **1820** includes upper gate **1822** between source/drain region **1824A** and source/drain region **1824B** and upper gate **1826** between source/drain region **1824B** and source/drain region **1824C**. Upper gates **1812**, **1816**, **1822**, **1826** may be poly gates or other types of gates for FET transistor devices. In one embodiment, upper gates **1812**, **1816**, **1822**, **1826** are NMOS gates.

(117) In certain embodiments, upper gate **1812** is separated from upper gate **1822** and upper gate **1816** is separated from upper gate **1826**. For instance, the poly for upper gate **1812** is not connected to the poly for upper gate **1822**. Similarly, the poly for upper gate **1816** is not connected to the poly for upper gate **1826**. The upper gates may be separated by either cutting the poly between the upper gates (e.g., cutting the poly between upper active region **1810** and upper active region **1820**) or forming the upper gates from separate poly layers in upper active region **1810** and upper active region **1820**. Separation of the upper gates between upper active region **1810** and upper active region **1820** provides distinction between transistors formed by these upper gates to allow the upper gates to form transistors for inverters and pass gates, as described herein.

(118) In the illustrated embodiments, lower active region **1830** includes lower gate **1832** between source/drain region **1834A** and source/drain region **1834B**. Lower active region **1840** includes lower gate **1846** between source/drain region **1844B** and source/drain region **1844C**. It should be noted that there are only two gate regions in the lower active regions **1830**, **1840** due to the presence of inactive portions **1833**, **1842** and as only two transistors are needed in combination with the four transistors in the upper active regions in order to form a memory cell device. Lower gates **1832**, **1846** may be poly gates or other types of gates for FET transistor devices. In one embodiment, lower gates **1832**, **1846** are PMOS gates.

(119) In certain embodiments, as shown in FIG. 21, portions of lower gate **1832** and lower gate **1846** extend across the separation distance **1835** between lower active region **1830** and lower active region **1840**. The portions of lower gate **1832** and lower gate **1846** extend across the separation distance are also seen in the depiction of FIG. 20 in the gap between upper active region **1810** and upper active region **1820**. In some embodiments, the portions of lower gate **1832** and lower gate **1846** extend across the separation distance **1835** extend below a transistor region of the other lower active region. For example, as shown in FIGS. 20 and 21, lower gate **1832** extends into a transistor region around lower active region **1840**, which is below the transistor region around upper active region **1820**. Similarly, lower gate **1846** extends into a transistor region around lower active region **1830**, which is below the transistor region around upper active region **1810**. In some embodiments, the portions of lower gate **1832** and lower gate **1846** extend under the transistor regions that define pass gates in the upper active regions, as described in more detail below. The extensions of lower gate **1832** and lower gate **1846** across the active regions provides capabilities for cross-coupling connections in cell **1800**, as also described in more detail below.

(120) An example embodiment of a 6T (six-transistor) SRAM memory cell that may be implemented in cell **1800** is now described with respect to the various connections made within the cell to implement the six transistors (e.g., four NMOS transistors and two PMOS transistors) that are arranged as inverters and pass gates. It should be understood that various additional embodiments of memory cells may be contemplated based on the disclosed structure of cell **1800**. As shown in FIG. 19, a 6T SRAM memory cell includes two NMOS transistors and two PMOS

transistors arranged to form two inverters, which are then cross-coupled. Two more NMOS transistors are then arranged to form pass gates connected to the inverters.

(121) Turning back to cell **1800**, upper gate **1812** along with source/drain region **1814A** and source/drain region **1814B**, shown in FIG. **20**, may form first NMOS transistor **1850** of inverter **1710A**. Then, as shown in FIG. **21**, lower gate **1832** along with source/drain region **1834A** and source/drain region **1834B** may form first PMOS transistor **1852** of inverter **1710A**. To form inverter **1710A** with transistor **1850** and transistor **1852**, source/drain region **1814A** is merged with source/drain region **1834A** by S/D merge **1860A**. S/D merge **1860A** may be, for example, a via or other substantially vertical connection made between source/drain region **1814A** and source/drain region **1834A**. Merging of source/drain region **1814A** and source/drain region **1834A** merges power connections between transistor **1850** and transistor **1852**.

(122) Additionally for transistor **1850** and transistor **1852**, source/drain region **1814B** is merged with source/drain region **1834B** by S/D merge **1860B** (as shown in FIG. **22**). Merging of source/drain region **1814B** and source/drain region **1834B** merges the outputs of transistor **1850** and transistor **1852**. The inputs of transistor **1850** and transistor **1852** may be merged by merging upper gate **1812** with lower gate **1832** using gate merge **1862A**. Gate merge **1862A**, as shown in FIG. **23**, may be a via or other substantially vertical connection made between upper gate **1812** and lower gate **1832**. With the inputs and outputs of transistor **1850** and transistor **1852** merged, the transistors form inverter **1710A**.

(123) Inverter **1710B**, shown in FIG. **19**, may similarly be formed by second NMOS transistor **1854** and second PMOS transistor **1856**, shown in FIGS. **20** and **21**. Transistor **1854** may be formed by upper gate **1826** along with source/drain region **1824B** and source/drain region **1824C**, as shown in FIG. **20**. Transistor **1856** may be formed by lower gate **1846** along with source/drain region **1844B** and source/drain region **1844C**, as shown in FIG. **21**. source/drain region **1814A** is merged with source/drain region **1834A** by S/D merge **1860A**.

(124) To form inverter **1710B** with transistor **1854** and transistor **1856**, source/drain region **1824B** is merged with source/drain region **1844B** by S/D merge **1860C** (also shown in FIG. **22**) and source/drain region **1824C** is merged with source/drain region **1844C** by S/D merge **1860D** (also shown in FIG. **24**). Merging of source/drain region **1824B** and source/drain region **1844B** merges the outputs of transistor **1854** and transistor **1856** while merging of source/drain region **1824C** and source/drain region **1844C** merges power connections between transistor **1854** and transistor **1856**. The inputs of transistor **1854** and transistor **1856** are then merged by merging upper gate **1826** with lower gate **1846** using gate merge **1862B**. With the inputs and outputs of transistor **1854** and transistor **1856** merged, the transistors form inverter **1710B**.

(125) In various embodiments, cell **1800** provides availability for forming pass gate **1720** and pass gate **1722** in addition to the inverters **1710A**, **1710B**. For instance, pass gate **1720** may be formed with third NMOS transistor **1858** while pass gate **1722** is formed with fourth NMOS transistor **1859**, as shown in FIG. **20**. It should be noted that both transistor **1858** and transistor **1859** are formed without any underlying PMOS transistor (e.g., above the inactive portions of the lower active regions). In the illustrated embodiment, transistor **1858** is formed by upper gate **1822** along with source/drain region **1824A** and source/drain region **1824B**. Transistor **1859** is formed by upper gate **1816** along with source/drain region **1814B** and source/drain region **1814C**.

(126) Both upper gate **1816** and upper gate **1822** may be coupled to a wordline (e.g. wordline **1730**) for the transmission of control signals to the gates. A read/write data connection to a bitline (e.g., bitline **1742**) for upper gate **1816** may be provided through source/drain region **1814C** while a read/write data connection to a bitline (e.g., bitline **1740**) for upper gate **1822** may be provided through source/drain region **1824A**. The output of transistor **1859** (which corresponds to pass gate **1722**) is provided through source/drain region **1814B**, which is also the output of transistor **1850** and merged with the output of transistor **1852** in inverter **1710A**. Correspondingly, the output of transistor **1858** (which corresponds to pass gate **1720**) is provided through source/drain region

1824B, which is also the output of transistor **1854** and merged with the output of transistor **1856** in inverter **1710B**. Thus, transistors **1858**, **1859** provide pass gate transistors **1720**, **1722** that are coupled to inverter **1710A** and inverter **1710B** according to the schematic diagram of FIG. **19**. (127) As discussed above, in certain embodiments, lower gate **1832** (in transistor **1852** of inverter **1710A**) and lower gate **1846** (in transistor **1856** of inverter **1710B**) extend towards the inactive portions of the active regions of the opposite transistor region. These extensions provide capability for providing cross-coupling between the inverters below the active regions of cell **1800**. For example, as shown in FIG. **23**, lower gate **1832** extends below gate **1822** formed in active region **1820**. With the extension of lower gate **1832**, cross-coupling **1864B** can be coupled between lower gate **1832** (which is the merged PMOS transistor gate in inverter **1710A**) and source/drain region **1844B** (which is the merged PMOS source/drain region in inverter **1710B**). Thus, cross-coupling **1864B** cross-couples the input of inverter **1710A** and the output of inverter **1710B**. Similarly, cross-coupling **1864A**, shown in FIGS. **20-22**, may be implemented to cross-couple the input of inverter **1710B** (by coupling to the extension of lower gate **1846**) and the output of inverter **1710A** (by coupling to source/drain region **1834B**, as shown in FIG. **22**).

(128) In certain embodiments, cross-couplings **1864A**, **1864B** are positioned below the active regions and above backside metal layers **120** in cell **1800**. For example, as shown in FIGS. **22** and **23**, cross-couplings **1864A**, **1864B** are coupled to the backside (e.g., bottom) of lower gates and source/drain regions in the lower transistor region of the PMOS transistors. Cross-couplings **1864A**, **1864B** may be placed in this area due to the removal of material in inactive portions **1833**, **1842** of lower active regions **1830**, **1840**. Cross-couplings **1864A**, **1864B**. The use of cross-couplings **1864A**, **1864B** in cell **1800** maintains a current design philosophy for a SRAM cell while reducing the cell height versus typical SRAM cells. For example, cross-couplings **1864A**, **1864B**, as implemented with lower gate **1832** and lower gate **1846**, provides better area scaling in cell **1800** by allowing both the upper active regions **1810**, **1820** and the lower active regions **1830**, **1840** to be brought closer together. For instance, in some embodiments, both the upper active regions **1810**, **1820** and the lower active regions **1830**, **1840** may be positioned with minimum required spacing between the diffusion regions in the active regions. Implementing the minimum required spacing may reduce the height of cell **1800** to about $\frac{1}{2}$ the typical height of a 6T SRAM cell.

(129) Macro SRAM Cells

(130) In various embodiments, in addition to being implemented in memory cells such as SRAM cells (e.g., SRAM bit cells), stacked transistors, such as those described herein, may be implemented in periphery cells associated with SRAM cells. For instance, the present disclosure contemplates various techniques that implement column input/output logic cells that include the stacked transistors. Implementation of stacked transistors in periphery cells, such as column input/output logic cells, may allow utilization of both top-side and backside routing in a memory device containing various types of SRAM cells, including the various embodiments of SRAM cells described herein.

(131) Certain embodiments disclosed herein have four broad elements: 1) a plurality of bit cells formed in first and second transistor regions that are vertically disposed relative to each other; 2) a first metal layer located above the bit cells (e.g., a top-side metal layer) and a second metal layer located below the bit cells (e.g., a backside metal layer), 3) a first column input/output logic cell coupled a first array of bit cells, and 4) a second column input/output logic cell coupled a second array of bit cells where the second array of bit cells is closer to the logic cells than the first array of bit cells. In certain embodiments, the first metal layer includes first routing that couples the first array of bit cells to the first column input/output logic cell while the second metal layer includes second routing that couples the second array of bit cells to the second column input/output logic cell. In some embodiments, the column input/output logic cells implement stacked transistors, such as those described herein.

(132) Accordingly, in various embodiments, the first column input/output logic cell provides

column I/O logic for bit cells that are further away from the periphery region of the device while the second column input/output logic cell provides column I/O logic for bit cells that are closer to the periphery region of the device. Splitting the routing between the topside and backside metal layers reduces routing congestion compared to utilization of only topside or backside routing for routing logic in a memory device. In various embodiments, dummy cells may be utilized for localized routing of bitline signals between topside and backside metal layers. For instance, dummy cells may be utilized for localized routing near bit cells of the first array (e.g., bit cells far away from the logic cells).

(133) In short, the present inventors have recognized that routing for column I/O logic in both topside metal layers and backside metal layers can be utilized in a memory device to relieve metal congestion in the memory device. Additionally, various techniques are implemented to reduce any area penalties associated with front to back transitions (or vice versa) when both topside metal layers and backside metal layers are used for routing logic between bit cells and logic cells. In some embodiments, the various routing paths provide reduced resistance paths for logic within a memory device. With implementation of the various disclosed techniques, the disclosed embodiments of a memory device may have strong signal connectivity with improved read/write speeds and thus improved performance.

(134) FIG. 25 depicts a block diagram representation of a memory device, according to some embodiments. In the illustrated embodiment, memory device **2300** includes memory cell region **2310** and logic circuit cell region **2320**. Memory cell region **2310** includes a plurality of bit cells that may be divided into far bit cell array **2312A** and near bit cell array **2312B**. The bit cells in the arrays may be, for example, SRAM cells **1800**, described above. Far bit cell array **2312A** includes a plurality of bit cells that are positioned further away from logic circuit cell region **2320** than the bit cells in near bit cell array **2312B**, as depicted in FIG. 25.

(135) In various embodiments, logic circuit cell region **2320** includes multiple column input/output (I/O) logic cells **2322**. Column I/O cells **2322** may, for instance, manage read/writes from bit cell arrays **2312**. Column I/O cells **2322** may also include portions of sense amplifiers. It should be understood that logic circuit cell region **2320** may include other logic cells in addition to column input/output (I/O) logic cells **2322**. For instance, logic circuit cell region **2320** may also include power switch logic cells, wordline logic circuit cells, local I/O circuit cells, global I/O circuit cells, etc. In some embodiments, logic circuit cell region **2320** may be referred to as a periphery region of memory device **2300**.

(136) In a contemplated embodiment, logic circuit cell region **2320** includes individual column I/O logic cells **2322** for each bit cell array in memory cell region **2310**. For instance, in the illustrated embodiment, logic circuit cell region **2320** includes first column I/O logic cell **2322A** and second column I/O logic cell **2322B** as memory cell region **2310** has two bit cell arrays **2312A**, **2312B**.

(137) In various embodiments, first column I/O logic cell **2322A** provides column I/O logic for far bit cell array **2312A** and second column I/O logic cell **2322B** provides column I/O logic for near bit cell array **2312B**. Routing in both topside metal layers **112** and backside metal layers **120** may be utilized in memory device **2300** to relieve metal congestion in the memory device. The present disclosure contemplates routing methods that also reduce any front to back transition area penalty in the utilization of both topside metal layers **112** and backside metal layers **120** in routing logic between bit cells and logic cells.

(138) In certain embodiments, memory device **2300** utilizes both topside metal layers **112** and backside metal layers **120** for bitline routing in the memory device. For example, topside metal layers **112** may be utilized for bitline routing between near bit cell array **2312B** and second column I/O logic cell **2322B** and backside metal layers **120** may be utilized for bitline routing between far bit cell array **2312A** and first column I/O logic cell **2322A**. In the illustrated embodiment, bitlines **1740A** and bitlines **1742A** provide bitline routing in far bit cell array **2312A** and bitlines **1740B** and bitlines **1742B** provide bitline routing in near bit cell array **2312B**. Bitlines **1740** and bitlines **1742**

may be, as described herein, complementary bitlines.

(139) As shown in FIG. 25, bitline **1740A** and bitline **1742A** in far bit cell array **2312A** are coupled to bitline output **2314A** and bitline output **2316A**, respectively. Bitline output **2314A** is then coupled to backside bitline routing **2330** and bitline output **2316A** is coupled to backside bitline routing **2332**. In certain embodiments, bitline **1740A** and bitline **1742A** are in topside metal layers **112**. For instance, as shown in FIGS. 19 and 20, the output of the pass gate transistors are in the upper transistor region and coupled to topside metal layers **112**. Because bitline **1740A** and bitline **1742A** are routing in the topside metal layers **112**, as shown in FIG. 25, a transition needs to be made from the topside metal layers to the backside metal layer **120** where backside bitline routing **2330** and backside bitline routing **2332** are positioned.

(140) In certain embodiments, dummy cells **2340A**, **2340B** are positioned at or near bitline outputs **2314A**, **2316A**, respectively. Dummy cell **2340A** includes a connection between bitline **1740A** in topside metal layers **112** and backside bitline routing **2330** in backside metal layers **120**. Dummy cell **2340B** includes a connection between bitline **1742A** in topside metal layers **112** and backside bitline routing **2332** in backside metal layers **120**. FIG. 26 depicts a topside plan view representation of a region having dummy cells **2340**, according to some embodiments. FIG. 27 depicts a backside plan view representation of the region having dummy cells **2340**, according to some embodiments. FIG. 28 depicts a cross-sectional representation of the region having dummy cells **2340** along the line A-A' shown in both FIGS. 26 and 27. FIG. 29 depicts a cross-sectional representation of the region having dummy cells **2340** along the line B-B' shown in both FIGS. 26 and 27.

(141) In various embodiments, as shown in FIG. 26, topside metal layers **112** in region **2400** includes routing for ground signals **2401** and wordlines **2402** in addition to bitline **1740A** and bitline **1742A**. In various embodiments, as shown in FIG. 27, backside metal layers **120** in region **2400** includes routing for signal **2500** and power signals **2502** in addition to backside bitline routing **2330** and backside bitline routing **2332**. In certain embodiments, dummy gates **2410** are positioned adjacent to active gates **2440** on both sides of region **2400**. Dummy gates **2410** may be, for example, gate cuts or other gates that isolate the area between the dummy gates. Isolation may include, for example, inhibiting connections to any gate activity within the area between the dummy gates **2410**.

(142) In certain embodiments, dummy cells **2340A**, **2340B** include trench metal **2420** formed between bitline **1740A** and backside bitline routing **2330** and between bitline **1742A** and backside bitline routing **2332**, as shown in FIGS. 26-29. Bitlines **1740A**, **1742A** may be coupled to trench metals **2420** by vias **2430** (shown in FIGS. 26 and 28-29) and backside bitline routings **2330**, **2332** may be coupled to trench metals **2420** by vias **2530** (shown in FIGS. 27-29).

(143) The use of trench metals **2420** for the connection between bitlines **1740A**, **1742A** and backside bitline routings **2330**, **2332** provides a low electrical resistance path for bitline signals to transition from topside metal layers **112** to backside metal layers **120**. Dummy cells **2340** provide localized traffic management for bitline signals at or near far bit cell array **2312A**. While dummy cells **2340** have some area penalty in memory device **2300**, the area penalty is small as the dummy cells are localized to shallow metal layers and are not associated with any global routing.

(144) Turning back to FIG. 25, after bitline signals are routed to backside bitline routing **2330** and backside bitline routing **2332** at dummy cells **2340A**, **2340B**, respectively, the backside bitline routing carries the signals to bitline input **2324A** and bitline input **2326A**, respectively, at first column I/O cell **2322A**. As described below, bitline input **2324A** and bitline input **2326A** in first column I/O cell **2322A** (and bitline inputs in second column I/O cell **2322B**) are inputs that are in the backside metal layers. Thus, no additional transition between topside metal layers **112** and backside metal layers **120** is needed for transmission of bitline signals from backside bitline routing **2330** and backside bitline routing **2332** to bitline input **2324A** and bitline input **2326A** in first column I/O cell **2322A**.

(145) Turning now to near bit cell array **2312B**, bitline **1740B** and bitline **1742B** are coupled to bitline output **2314B** and bitline output **2316B**, respectively. Bitline output **2314B** and bitline output **2316B** are then coupled to topside bitline routing **2334** and topside bitline routing **2336**, respectively. Since both bitlines **1740B**, **1742B** and topside bitline routings **2334**, **2336** are located in topside metal layers **112**, no transition between the topside metal layers and backside metal layers **120** is needed at near bit cell array **2312B**.

(146) In the illustrated embodiment, topside bitline routing **2334** and topside bitline routing **2336** carry bitline signals from near bit cell array **2312B** to bitline input **2324B** and bitline input **2326B**, respectively, at second column I/O cell **2322B**. As noted above and described below, bitline input **2324B** and bitline input **2326B** are positioned in backside metal layers **120**. Accordingly, a transition from topside metal layers **112** to backside metal layers **120** may be needed at bitline input **2324B** and bitline input **2326B**.

(147) In certain embodiments, dummy cell **2350** is positioned at or near bitline inputs **2324B**, **2326B**. Dummy cell **2350** includes a connection between topside bitline routing **2334** in topside metal layers **112** and bitline input **2324B** in backside metal layers **120** and a connection between topside bitline routing **2336** in topside metal layers **112** and bitline input **2326B** in backside metal layers **120**. FIG. **30** depicts a topside plan view representation of a region having dummy cell **2350**, according to some embodiments. FIG. **31** depicts a backside plan view representation of the region having dummy cell **2350**, according to some embodiments. FIG. **32** depicts a cross-sectional representation of the region having dummy cell **2350** along the line A-A' shown in both FIGS. **30** and **31**. FIG. **33** depicts a cross-sectional representation of the region having dummy cell **2350** along the line B-B' shown in both FIGS. **30** and **31**.

(148) In various embodiments, as shown in FIG. **30**, topside metal layers **112** in region **2800** includes routing for ground signal **2802** and signals **2804** in addition to topside bitline routing **2334** and topside bitline routing **2336**. In various embodiments, as shown in FIG. **31**, backside metal layers **120** in region **2800** includes routing for signal **2900** and power signal **2902** in addition to bitline input **2324B** and bitline input **2326B**. In certain embodiments, dummy gates **2810** are positioned adjacent to active gates **2840** on both sides of region **2800**. Dummy gates **2810** may be, for example, gate cuts or other gates that isolate the area between the dummy gates. Isolation may include, for example, inhibiting connections to any gate activity within the area between the dummy gates **2810**.

(149) In certain embodiments, dummy cell **2350** includes trench metal **2820** formed between topside bitline routing **2334** and bitline input **2324B** and between topside bitline routing **2336** and bitline input **2326B**, as shown in FIGS. **30-33**. Topside bitline routings **2334**, **2336** may be coupled to trench metals **2820** by vias **2830** (shown in FIGS. **30** and **32-33**) and bitline inputs **2324B**, **2326B** may be coupled to trench metals **2820** by vias **2930** (shown in FIGS. **31-33**).

(150) Similar to dummy cells **2340**, the use of trench metals **2820** for the connection between topside bitline routings **2334**, **2336** and backside bitline inputs **2324B**, **2326B** provides a low electrical resistance path for bitline signals to transition from topside metal layers **112** to backside metal layers **120**. Dummy cells **2350** provide localized traffic management for bitline signals in logic circuit cell region **2320** (e.g., in the periphery region).

(151) Turning back to FIG. **25**, as with first column I/O cell **2322A**, bitline inputs **2324B**, **2326B** in second column I/O cell **2322B** are inputs that are in backside metal layers **120**. Thus, after bitline signals are routed to bitline input **2324B** and bitline input **2326B** by dummy cell **2350**, second column I/O cell **2322B** can receive bitline signals in the proper metal layer. With the various routings of bitline signals from memory cell region **2310** to logic circuit cell region **2320** through combinations of topside metal layers **112** and backside metal layers, column logic I/O cells **2322** in the logic circuit cell region may, in various embodiments, have unipolar connectivity in a simple fabrication scheme.

(152) In various embodiments, column I/O cells **2322** may implement stacked transistors to

provide connectivity to the various routings described above for memory device **2300**. FIG. **34** depicts a schematic representation of a column I/O cell **2322**, according to some embodiments. In the illustrated embodiment, cell **2322** includes five PMOS transistors and two NMOS transistors. The PMOS transistors include PMOS1 transistor **3210**, PMOS2 transistor **3220**, PMOS3 transistor **3230**, PMOS4 transistor **3240**, and PMOS5 transistor **3250**. The NMOS transistors includes NMOS1 transistor **3260** and NMOS2 transistor **3270**.

(153) As shown in FIG. **34**, various routings and connections for the transistors in cell **2322** are provided by topside metal layers **112** (solid lines) and backside metal layers **120** (dashed lines). In certain embodiments, cell **2322** includes Vdd **3202**, Vss **3203**, PCH **3204**, Rcs **3206**, Wcs **3208**, wordline outputs **3280A**, **3280B**, and sense outputs **3282A**, **3282B**. Vdd **3202** provides routing to power for cell **2322** while Vss **3203** provides routing to ground. PCH **3204** couples PMOS1 transistor **3210**, PMOS2 transistor **3220**, and PMOS3 transistor **3230** to form a precharge circuit. Rcs **3206** couples PMOS4 transistor **3240** and PMOS5 transistor **3250** for a read column select circuit and Wcs **3208** couples NMOS1 transistor **3260** and NMOS2 transistor **3270** for a write column select circuit. Wordline outputs **3280A**, **3280B** provide write outputs from cell **2322** and sense outputs **3282A**, **3282B** provide read outputs from cell **2322**.

(154) In certain embodiments, the transistors in cell **2322** may be formed by a stacked transistor as described herein. For example, the PMOS transistors are formed in a lower transistor region and the NMOS transistors are formed in an upper transistor region. FIG. **35** depicts a layout of cell **2322**, according to some embodiments. In FIG. **35**, the top panel is a topside plan view representation of the upper transistor region **3300** and the bottom panel is a backside plan view representation of the lower transistor region **3350**. Upper transistor region **3300** includes upper active region **3302** and lower transistor region **3350** includes lower active region **3352**.

(155) In the illustrated embodiment of upper transistor region **3300** in FIG. **35**, topside metal layers **112** includes routing for bitline input **2324**, bitline input **2326**, Vss **3203**, Wcs **3208**, and wordline outputs **3280A**, **3280B**. NMOS1 transistor **3260** includes by gates **3262A**, **3262B**, **3262C** and source/drain regions **3264A**, **3264B**, **3264C**, **3264D**. NMOS2 transistor **3270** includes by gates **3272A**, **3272B**, **3272C** and source/drain regions **3274A**, **3274B**, **3274C**, **3274D**. Dummy gate **3310** (e.g., a gate cut) separates NMOS1 transistor **3260** and NMOS2 transistor **3270**. Vias **3312** connect gates **3262** and gates **3272** using Wcs **3208**. Vias **3312** also connect source/drain region **3264A** and source/drain region **3264C** to wordline output **3280A** along with source/drain region **3274B** and source/drain region **3274D** to wordline output **3280B**. Yet more vias **3312** connect source/drain region **3264B** and source/drain region **3264D** to bitline input **2324** and source/drain region **3274A** and source/drain region **3274C** to bitline input **2326**.

(156) In the illustrated embodiment of lower transistor region **3350** in FIG. **35**, backside metal layers **120** includes routing for bitline input **2324**, bitline input **2326**, Vdd **3202**, PCH **3204**, Rcs **3206**, and sense outputs **3282A**, **3282B**. PMOS1 transistor **3210** includes by gate **3212** and source/drain regions **3214A**, **3214B**. PMOS2 transistor **3220** includes by gate **3222** and source/drain regions **3214B**, **3214C**. PMOS2 transistor **3230** includes by gate **3232** and source/drain regions **3214C**, **3214D**. Dummy gate **3320** then separates source/drain region **3214D** and PMOS2 transistor **3230** from source drain/region **3244A** in PMOS4 transistor **3240**. PMOS4 transistor **3240** includes gate **3242** and source/drain regions **3244A**, **3244B**. Another dummy gate **3320** then separates PMOS4 transistor **3240** and PMOS5 transistor **3250**. PMOS5 transistor **3250** includes gate **3252** and source/drain regions **3254A**, **3254B**.

(157) Vias **3322** connect gate **3212**, gate **3222**, and gate **3232** by PCH **3204**. Vias **3322** also connect gate **3242** and **3252** using Rcs **3206**. More vias **3322** connect source/drain region **3244B** to sense output **3282A** and source/drain region **3254B** to sense output **3282B**. Yet more vias **3322** connect source/drain region **3214B** and source/drain region **3244A** to bitline input **2324** and source/drain region **3214C** and source/drain region **3254A** to bitline input **2326**. Vdd is connected to source/drain region **3214A** and source/drain region **3214D** by additional vias **3322**.

(158) In some embodiments, source/drain region **3214B** in lower transistor region **3350** is merged with source/drain region **3264B** in upper transistor region **3300** by source/drain merge **3290A**. Additionally, source/drain region **3254A** may be merged with source/drain region **3274C** by source/drain merge **3290B**. Merging of these source/drain regions provides necessary connections between NMOS transistors and PMOS transistors.

(159) The embodiment of memory device **2300** described herein provides a memory device that may provide strong signal connectivity using current layout techniques without significant area penalties. Routing in memory device **2300** utilizes bitline routing through topside and backside metal layers to avoid metal routing congestion in the device. Memory device **2300** also avoids typical complexities involved with unipolar device fabrication utilizing the various routing paths and connection paths described herein. The various connection paths described herein may also reduce resistance within memory device **2300** and thus improve read/writing speeds and the performance of the memory device.

(160) Vertical Transistor Cells

(161) A recent development in transistor design is the implementation of vertical transistors where the cells have vertical transport through vertically displaced source/drain regions and a gate positioned vertically in between the source/drain regions. Current vertical transistor designs typically include wide frontside (e.g., topside) power rails at the boundaries of the cell for power delivery. These wide power rails, however, contribute to an increased and large standard cell height. The larger standard cell height reduces the area efficiency of the vertical transistor while also reducing available connectivity and performance of the transistor.

(162) The present disclosure contemplates various embodiments that utilize backside power routing in vertical transistor designs to reduce scaling, provide better connectivity, and provide better performance of the transistors. Certain embodiments disclosed herein have four broad elements: 1) a pair of vertical transistors in an integrated circuit cell; 2) a topside metal layers above the transistor regions of the vertical transistors with signal routing, 3) a backside metal layer below the transistor regions with power routing, and 4) a metal contact layer between the backside metal layer and source/drain regions of the transistors. In certain embodiments, the transistors are complementary transistors. In some embodiments, vias couple power routing in the backside metal layer to the metal contact layer. In some embodiments, a second pair of vertical transistors may be included in the cell. Additional implementations of gate vias, fins, contact vias, and various other connections and routings may also be contemplated in various embodiments.

(163) In various embodiments, control signal and power signal connections are made using various contacts or vias to implement logic associated with specific integrated circuit devices having multiple vertical transistors for the cell constructions described herein. For instance, examples of an inverter device, a NAND device, and a MUX device that may be implemented based on the vertical transistor cell construction are described below. Embodiments of various possible connections for control signals and voltage signals to the vertical transistors within the cell constructions are also described. A person with knowledge in the art would understand that combinations of these various possible connections may be implemented to generate many different desired circuits based on the vertical transistor structure within the cell constructions disclosed herein.

(164) In short, the present inventors have recognized that the implementation of backside routing for power connections in combination with vertical transistors provides various opportunities for construction of specific transistor designs with reduced scaling. Additionally, various techniques are implemented to provide specific routing for control signal and power routing within the cell constructions with vertical transistors described herein. With implementation of the various disclosed techniques, vertical transistor cell constructions that provide improved performance in a small scale factor are contemplated.

(165) FIG. **36** depicts a perspective representation of a contemplated vertical transistor device, according to some embodiments. FIG. **37** depicts a perspective representation of another

contemplated vertical transistor device, according to some embodiments. It should be noted that device **3400**, shown in FIG. **36**, and device **3500**, shown in FIG. **37**, are generic representations of vertical transistor-based device structures without depiction of various connections that can be made to the structures. Example embodiments of connected structures are further disclosed herein below with respect to FIGS. **38-54**.

(166) In the illustrated embodiment of FIG. **36**, device **3400** includes two vertical transistors **3410**, **3420**. In certain embodiments, transistors **3410**, **3420** are complementary types of transistors. For instance, transistor **3410** is a PMOS transistor and transistor **3420** is an NMOS transistor. Transistor **3410** includes lower source/drain region **3412**, gate **3414**, and upper source/drain region **3416**. Similarly, transistor **3420** includes lower source/drain region **3422**, gate **3424**, and upper source/drain region **3426**. In some embodiments, gate **3414** and gate **3424** are fin-type gates. In various embodiments, gate **3414** includes gate spacers **3415** and gate **3424** includes gate spacers **3425**. Gate spacers **3415**, **3425** are not labelled in the remaining figures for simplicity in the drawings.

(167) As depicted in FIG. **36**, the lower source/drain regions, the gates, and the upper source/drain regions are stacked in the vertical dimension of the transistors. Further as depicted, transistor **3410** and transistor **3420** are parallel and have a spacing (e.g., distance) between them in the horizontal direction (e.g., the horizontal dimension) of device **3400**.

(168) In certain embodiments, transistor **3410** includes upper contact **3418** coupled to upper source/drain region **3416** and transistor **3420** includes upper contact **3428** coupled to upper source/drain region **3426**. Contact **3418** and contact **3428** may be, for example, metal contacts for contacting various resources in a first metal layer positioned above transistor **3410** and transistor **3420**. For example, as shown in FIG. **36**, contact **3418** may be routed to a resource by route **3430** (e.g., a routing shown by the dotted line). Route **3430** may be, for example, a metal layer route path in a first metal layer above transistor **3410** and transistor **3420**. It should be noted that the dotted line depiction of route **3430** is provided as example of one resource (e.g., routing) in the metal layer and that the metal layer may include multiple resources (e.g., multiple routings). Additionally, only the first metal layer above transistor **3410** and transistor **3420** is depicted and there may be multiple additional metal routing above route **3430**.

(169) In various embodiments, transistor **3410** includes lower contact **3419** coupled to lower source/drain region **3412** and transistor **3420** includes lower contact **3429** coupled to lower source/drain region **3422**. Contacts **3419**, **3429** may be, for example, metal contacts. Contacts **3419**, **3429** may be utilized to route to backside power routing layers (e.g., backside power routing **3440A** or backside power routing **3440B**, as shown in FIG. **36** and described herein) or to route to various other resources within device **3400**.

(170) In certain embodiments, device **3400** includes a backside power layer. In the illustrated embodiment of FIG. **36**, the backside power layer includes backside power routing **3440A** and backside power routing **3440B**. Routing **3440A** and routing **3440B** may, for example, provided routing to/from power source (e.g., Vdd) and power ground (e.g., Vss) resources for device **3400**.

(171) In various embodiments, gate **3414** and gate **3424** are interconnected by gate bridge **3450**. Gate bridge **3450** may be formed, for example, by extension of the gate material of gate **3414** and gate **3424** to couple the gates together. In some embodiments, gate bridge **3450** may be formed by a single extension of gate material from either gate **3414** or gate **3424** that is extended to the other gate. Gate bridge **3450** may also include extension of material for gate spacers. Gate bridge **3450** merges gate **3414** and gate **3424** for implementation of transistor **3410** and transistor **3420** in various embodiments of CMOS devices, some examples of which are described herein. Various embodiments may also be contemplated where gate **3414** and/or gate **3424** extend in other directions. For instance, a gate may include an extension that extends towards an outer boundary of device **3400** (e.g., towards an outer boundary of the cell structure in an opposite direction of gate bridge **3450**).

(172) In the illustrated embodiment of FIG. 37, device 3500 does not have a gate bridge connecting gate 3414 in transistor 3410 and gate 3424 in transistor 3420. Various techniques for connecting transistor 3410 and transistor 3420 may be contemplated without the gate bridge. For instance, in one contemplated embodiment, contact 3418 and contact 3428 may be connected by strap 3510. Strap 3510 may be, for example, a metal strap. In some embodiments, contact 3418, contact 3428, and strap 3510 may be formed as a single contact (e.g., a single strap connecting upper source/drain region 3416 and upper source/drain region 3426). Various embodiments may also be contemplated where strap 3510 extends in another direction from one of contacts 3418, 3428. For instance, strap 3510 may extend perpendicular to the depicted embodiment towards another vertical transistor or resource in device 3500.

(173) In another contemplated embodiment, contact 3419 and contact 3429 may be connected by strap 3520. Strap 3520 may also be a metal strap. In some embodiments, strap 3520 is formed as a single contact along with contact 3419 and contact 3429. For example, strap 3520, contact 3419, and contact 3429 may be part of a single metal contact plate formed in the contact layer. Various embodiments may also be contemplated where contact 3419 and/or contact 3429 extends outwards from the bottoms of transistors 3410, 3420. For instance, a contact may have a portion that extends towards an outer boundary of device 3500 (e.g., towards an outer boundary of the cell structure).

(174) It should be understood that while device 3400, shown in FIG. 36, and device 3500, shown in FIG. 37, are depicted with various connection structures separately, embodiments may be contemplated where structures from device 3400 are combined with structures from device 3500 in a cell design. For instance, a device may be contemplated that includes both gate bridge 3450 and one or both of strap 3510 and strap 3520. Various example device cell constructions are now described as example based on device 3400 and/or device 3500. It should be noted that the various device cell constructions are provided as example and that various additional device cell constructions may be implemented based on the description herein.

(175) FIGS. 38-42 depict representations of an inverter cell construction, according to some embodiments. FIG. 38 depicts a perspective view representation of the inverter cell construction, according to some embodiments. FIG. 39 depicts a topside plan view representation of the inverter cell construction, according to some embodiments. FIG. 40 depicts a backside plan view representation of the inverter cell construction, according to some embodiments. FIG. 41 depicts a cross-sectional representation of the inverter cell construction, according to some embodiments, along line 41-41 shown in FIG. 39 (e.g., along the gate bridge). FIG. 42 depicts a cross-sectional representation of the inverter cell construction, according to some embodiments, along line 42-42 shown in FIG. 39 (e.g., perpendicular to the gate fin of transistor 3410).

(176) Inverter cell device 3600 may be derived from the structure of device 3400, shown in FIG. 36. In the illustrated embodiment of FIGS. 38-42, device 3600 includes vertical transistor 3410 and vertical transistor 3420. Transistor 3410 includes lower source/drain region 3412, gate 3414, upper source/drain region 3416, upper contact 3418, and lower contact 3419. Transistor 3420 includes lower source/drain region 3422, gate 3424, upper source/drain region 3426, upper contact 3428, and lower contact 3429. In the illustrated embodiment of device 3600, transistor 3410 is a PMOS transistor and transistor 3420 is an NMOS transistor.

(177) In certain embodiments, device 3600 includes backside vias 3610A, 3610B. Backside via 3610A is coupled to lower source/drain region 3412 through lower contact 3419. Backside via 3610A couples lower source/drain region 3412 to backside power routing 3440A. For device 3600, backside power routing 3440A provides power supply (e.g., V_{dd}) to lower source/drain region 3412 and transistor 3410. Backside via 3610B is coupled to lower source/drain region 3422 through lower contact 3429. Backside via 3610B couples lower source/drain region 3422 to backside power routing 3440B. For device 3600, backside power routing 3440B provides ground supply (e.g., V_{ss}) to lower source/drain region 3422 and transistor 3420.

(178) In various embodiments, device 3600 includes topside vias 3620A, 3620B. Topside via

3620A may be coupled to upper source/drain region **3416** through upper contact **3418** and top-side via **3620B** may be coupled to upper source/drain region **3426** through upper contact **3428**. Top-side vias **3620A**, **3620B** may provide connection to signal routing resources (e.g., routes **3430A-E**) in a first metal layer above transistor **3410** and transistor **3420**. For example, in the illustrated embodiment, top-side via **3620A** is coupled to route **3430B** and top-side via **3620B** is coupled to route **3430D**. Routes **3430B** and **3430D** may provide routes for output signals from transistor **3410** and transistor **3420**, respectively.

(179) In certain embodiments, a route for an input signal to transistor **3410** and transistor **3420** is provided by route **3430C**. As shown in FIGS. **38** and **39**, route **3430C** is coupled to gate via **3630**, which is coupled to gate bridge **3450**. Thus, gate via **3630** provides connection between route **3430C** (e.g., the input signal route) and both gate **3414** in transistor **3410** and gate **3424** in transistor **3420**. With the connections to the input signal route, the output signal routes, and the power supply/ground routes, transistor **3410** and transistor **3420** are connected to form the inverter cell device **3600**.

(180) It should be noted that while FIGS. **38** and **39** depict five routes **3430A-E** in the first metal layer above transistor **3410** and transistor **3420**, the first metal layer may include additional routes. Further, additional metal layers may be positioned above the first metal layer and provide various connections to either the first metal layer or device **3600**. For example, in one embodiment, a metal layer above the first metal layer may include a strap (or other connector) coupling route **3430B** and route **3430D** such that the outputs of transistor **3410** and transistor **3420** are merged together into a single output. Additionally, while two backside power routings are shown (e.g., routing **3440A** and routing **3440B**), the backside power layer may include additional routings (e.g., routings for other power and signal resources).

(181) The top-side and backside plan views of device **3600** shown in FIGS. **39** and **40** further depict gate fins that may be present in the gates of the transistors. For example, gate fin **3415** is the gate fin for gate **3414** and gate fin **3425** is the gate fin for gate **3424**. Gate fin **3415** and gate fin **3425** are also shown in the cross-sectional representation of device **3600** in FIG. **41** and gate fin **3415** is shown in the cross-sectional representation of transistor **3410** in FIG. **42**. Note that the cross-section representation of FIG. **42** is perpendicular to the gate fin of transistor **3410**, which is the direction of route **3430B**, shown in FIGS. **38** and **39**.

(182) FIGS. **43-47** depict representations of a NAND cell construction, according to some embodiments. FIG. **43** depicts a perspective view representation of the NAND cell construction, according to some embodiments. FIG. **44** depicts a top-side plan view representation of the NAND cell construction, according to some embodiments. FIG. **45** depicts a backside plan view representation of the NAND cell construction, according to some embodiments. FIG. **46** depicts a cross-sectional representation of the NAND cell construction, according to some embodiments, along line **46-46** shown in FIG. **44** (e.g., along gate bridge **3450'**). FIG. **47** depicts a cross-sectional representation of the NAND cell construction, according to some embodiments, along line **47-47** shown in FIG. **44** (e.g., perpendicular to the gate fins of transistor **3410** and transistor **3410'**).

(183) NAND cell device **4100** may be derived from the structure of device **3400**, shown in FIG. **36**. In the illustrated embodiment of FIGS. **43-47**, device **4100** includes vertical transistor **3410**, vertical transistor **3420**, vertical transistor **3410'**, and vertical transistor **3420'**. Transistor **3410** includes lower source/drain region **3412**, gate **3414**, and upper source/drain region **3416**. Transistor **3420** includes lower source/drain region **3422**, gate **3424**, and upper source/drain region **3426**. Transistor **3410'** includes lower source/drain region **3412'**, gate **3414'**, and upper source/drain region **3416'**. Transistor **3420'** includes lower source/drain region **3422'**, gate **3424'**, and upper source/drain region **3426'**. In the illustrated embodiment of device **4100**, transistor **3410** and transistor **3410'** are PMOS transistors and transistor **3420** and transistor **3420'** are NMOS transistors.

(184) In certain embodiments, route for input signals to transistor **3410**, transistor **3410'**, transistor

3420, and transistor **3420'** are provided by route **3430C**. As shown in FIGS. **43** and **44**, route **3430C** is coupled to gate via **3630A**, which is coupled to gate bridge **3450**, and gate via **3630B**, which is coupled to gate bridge **3450'**. Thus, gate via **3630A** provides connection between route **3430C** (e.g., the input signal route) and both gate **3414** in transistor **3410** and gate **3424** in transistor **3420**. Gate via **3630B** provides connection between route **3430C** (e.g., the input signal route) and both gate **3414'** in transistor **3410'** and gate **3424'** in transistor **3420'**.

(185) In certain embodiments, upper source/drain region **3416** of transistor **3410** and upper source/drain region **3416'** of transistor **3410'** are connected by contact **3418**. Similarly, upper source/drain region **3426** of transistor **3420** and upper source/drain region **3426'** of transistor **3420'** are connected by contact **3428**. In various embodiments, device **4100** includes topside via **3620** connected to contact **3418**. Topside via **3620** may provide connection to route **3430B** in the first metal layer above the transistor region of device **4100**. In the illustrated embodiment, route **3430B** provides a route for output signals from transistor **3410** and transistor **3410'**.

(186) In the illustrated embodiment, only transistor **3410**, transistor **3410'**, and transistor **3420** are connected to backside layers. For instance, transistor **3410** is connected to backside power routing **3440A** by contact **3419** and backside via **3610A**, transistor **3410'** is connected to backside power routing **3440A** by contact **3419'** and backside via **3610A'**, and transistor **3420** is connected to backside power routing **3440B** by contact **3429** and backside via **3610B**, as shown in FIGS. **42** and **45**. In various embodiments of device **4100**, backside power routing **3440A** provides power supply (e.g., V_{dd}) to lower source/drain region **3412** and transistor **3410** and to lower source/drain region **3412'** and transistor **3410'** while backside power routing **3440B** provides ground supply (e.g., V_{ss}) to lower source/drain region **3422** and transistor **3420**.

(187) In certain embodiments, lower source/drain region **3422'** in transistor **3420'** is connected to contact **3429'**, which is not connected to a backside power routing layer. Contact **3429'** extends away from lower source drain region **3422'** and towards a boundary of the cell, as shown in FIGS. **43**, **45**, and **46**. Contact **3429'** is then coupled to route **3430E** by contact via **4110**. Route **3430E** is a route in the first metal layer above the transistor region. Contact via **4110** is a via that belongs to the cell structure of device **4100** and is not shared with any neighboring cells along the cell boundary. In certain embodiments, route **3430E** is a signal route in the first metal layer for signal output from transistor **3420'**. Thus, a signal in the NMOS transistors (e.g., transistor **3420** and transistor **3420'**) routes from lower source/drain region **3422** (connected to ground by backside power routing **3440B**), through the transistors, and out through contact via **4110** to route **3430E**.

(188) In the illustrated embodiment, route **3430E** provides a route for output signals from transistor **3420** and transistor **3420'**. The output signals routed through route **3430E** may be combined with output signals from route **3430B**. For example, a metal layer above the first metal layer may include a strap (or other connector) coupling route **3430B** and route **3430E** such that the outputs of the transistors are merged together into a single output.

(189) The various routings and connections in device **4100** form the NAND cell device. FIGS. **44** and **45** illustrate gate fins **3415**, **3415'**, **3425**, **3425'** in gates **3414**, **3414'**, **3424**, **3424'**, respectively. Gate fins **3415'** and gate fin **3425'** are also shown in the cross-sectional representation of device **4100** in FIG. **46** and gate fin **3415** and gate fin **3415'** are shown in the cross-sectional representation of device **4100** in FIG. **47**. Note that the cross-section representation of FIG. **47** is perpendicular to the gate fins of transistor **3410** and transistor **3410'**, which is the direction of route **3430B**, shown in FIG. **44**.

(190) FIGS. **48-52** depict representations of a MUX (multiplexer) cell construction, according to some embodiments. FIG. **48** depicts a perspective view representation of the MUX cell construction, according to some embodiments. FIG. **49** depicts a topside plan view representation of the MUX cell construction, according to some embodiments. FIG. **50** depicts a backside plan view representation of the MUX cell construction, according to some embodiments. FIG. **51** depicts a cross-sectional representation of the MUX cell construction, according to some

embodiments, along line 51-51 shown in FIG. 49 (e.g., along gate fin 3415' and gate fin 3425"). FIG. 52 depicts a cross-sectional representation of the MUX cell construction, according to some embodiments, along line 52-52 shown in FIG. 49 (e.g., perpendicular to the gate fins of transistor 3410 and transistor 3410").

(191) MUX cell device 4600 may be derived from the structure of device 3500, shown in FIG. 37. In the illustrated embodiment of FIGS. 48-52, device 4600 includes vertical transistor 3410, vertical transistor 3420, vertical transistor 3410", and vertical transistor 3420". As in device 3500, there are gate bridges between gates of the transistors in device 4600 so that there are no common gates between complementary type transistors. Transistor 3410 includes lower source/drain region 3412, gate 3414, and upper source/drain region 3416. Transistor 3420 includes lower source/drain region 3422, gate 3424, and upper source/drain region 3426. Transistor 3410" includes lower source/drain region 3412", gate 3414", and upper source/drain region 3416". Transistor 3420" includes lower source/drain region 3422", gate 3424", and upper source/drain region 3426". In the illustrated embodiment of device 4600, transistor 3410 and transistor 3410" are PMOS transistors and transistor 3420 and transistor 3420" are NMOS transistors.

(192) As MUX cell device 4600 is a transmission device, none of transistor 3410 and transistor 3410" and none of transistor 3420 and transistor 3420" are connected to any power in the MUX cell structure. In various embodiments of MUX cell device 4600, the lower source/drain regions of the transistors are connected together (e.g., merged together). For instance, in the illustrated embodiment, contact plate 4620 is connected to lower source/drain region 3412 in transistor 3410, lower source/drain region 3412" in transistor 3410", lower source/drain region 3422 in transistor 3420, and lower source/drain region 3422" in transistor 3420".

(193) In certain embodiments, contact via 4630 is coupled to contact plate 4620. Contact via 4630 may be connected to contact plate 4620 at or near a center of the contact plate. Contact via 4630 then connects to route 3430C in the first metal layer above the transistor region. In various embodiments, route 3430C provides output routing for MUX cell device 4600. Thus, contact via 4630 may be referred to as an output pin of MUX cell device 4600.

(194) In various embodiments, gates 3414, 3414", 3424, 3424" are extended towards the boundary of the cell to provide surfaces for direct vertical connections to the gates from routes in the first metal layer above. For example, as illustrated in FIGS. 48-52, gate 3414 includes gate extension 4640A that extends toward the boundary of the cell (e.g., extends horizontally towards the boundary of the cell). Similarly, gate 3414" includes gate extension 4640B, gate 3424 includes gate extension 4640C, and gate 3424" includes gate extension 4640D. The gate extensions 4640A-D are then connected to routes in the first metal layer above by gate vias 3630A-D, respectively. For example, as shown in FIGS. 48 and 49, gate via 3630A connects gate extension 4640A to route 3430A, gate via 3630B connects gate extension 4640B to route 3430A, gate via 3630C connects gate extension 4640C to route 3430E, and gate via 3630D connects gate extension 4640D to route 3430E. One or both of route 3430A and route 3430E are located at the boundary of the cell and are not shared with neighboring cells. Route 3430A and route 3430E may provide input routes to device 4600.

(195) In certain embodiments, upper source/drain region 3416 in transistor 3410 is connected to upper source/drain region 3426 in transistor 3420 by contact 4610A. This connection merges upper source/drain region 3416 with upper source/drain region 3426. Similarly, upper source/drain region 3416" in transistor 3410" is connected to upper source/drain region 3426" in transistor 3420" by contact 4610B. With the merging of these upper source/drain regions and the common connection between the lower source/drain regions (and single output through contact via 4630), device 4600 may operate as a MUX (multiplexer) where signals are input through gate vias 3630A-D and output through contact via 4630.

(196) FIGS. 49 and 50 illustrate gate fins 3415, 3415", 3425, 3425" in gates 3414, 3414", 3424, 3424", respectively. Gate fins 3415 and gate fin 3425 are also shown in the cross-sectional

representation of device **4600** in FIG. **51** and gate fin **3415** and gate fin **3415'** are shown in the cross-sectional representation of device **4600** in FIG. **52**. Note that the cross-section representation of FIG. **52** is perpendicular to the gate fins of transistor **3410** and transistor **3410''**, which is the direction of route **3430B**, shown in FIG. **49**.

(197) FIGS. **53** and **54** depict representations of a cell device having dielectric walls, according to some embodiments. FIG. **53** depicts a perspective view representation of device **5100**, according to some embodiments. FIG. **54** depicts a cross-sectional representation of device **5100**, according to some embodiments, along line **54-54** shown in FIG. **53** (e.g., along gate bridge **3450'**).

(198) Device **5100** may be derived from the structure of device **3400**, shown in FIG. **36**. In some embodiments, device **5100** may be similar to the inverter cell device **4100**, shown in FIGS. **43-47**. In the illustrated embodiment of FIGS. **53** and **54**, device **5100** includes vertical transistor **3410** and vertical transistor **3420**. Transistor **3410** includes lower source/drain region **3412**, gate **3414**, and upper source/drain region **3416**. Transistor **3420** includes lower source/drain region **3422**, gate **3424**, and upper source/drain region **3426**. In certain embodiments, transistor **3410** is a PMOS transistor and transistor **3420** is an NMOS transistor.

(199) In various embodiments, as shown in FIGS. **53** and **54**, wall **5100A** may be positioned on one a first side of the cell (e.g., on a side of transistor **3410**) and wall **5100B** may be positioned on a second side of the cell (e.g., on a side of transistor **3420** opposite transistor **3410**). In certain embodiments, wall **5100A** and wall **5100B** are dielectric walls. Placing dielectric walls on one or both sides of device **5100** may reduce the space needed between device **5100** and another neighboring cell. Accordingly, wall **5100A** and wall **5100B** may be implemented when reducing in scaling of devices is necessary.

(200) Example Computer System

(201) Turning next to FIG. **55**, a block diagram of one embodiment of a system **5300** is shown that may incorporate and/or otherwise utilize the methods and mechanisms described herein. In the illustrated embodiment, the system **5300** includes at least one instance of a system on chip (SoC) **5306** which may include multiple types of processing units, such as a central processing unit (CPU), a graphics processing unit (GPU), or otherwise, a communication fabric, and interfaces to memories and input/output devices. In some embodiments, one or more processors in SoC **5306** includes multiple execution lanes and an instruction issue queue. In various embodiments, SoC **5306** is coupled to external memory **5302**, peripherals **5304**, and power supply **5308**.

(202) A power supply **5308** is also provided which supplies the supply voltages to SoC **5306** as well as one or more supply voltages to the memory **5302** and/or the peripherals **5304**. In various embodiments, power supply **5308** represents a battery (e.g., a rechargeable battery in a smart phone, laptop or tablet computer, or other device). In some embodiments, more than one instance of SoC **5306** is included (and more than one external memory **5302** is included as well).

(203) The memory **5302** is any type of memory, such as dynamic random access memory (DRAM), synchronous DRAM (SDRAM), double data rate (DDR, DDR2, DDR3, etc.) SDRAM (including mobile versions of the SDRAMs such as mDDR3, etc., and/or low power versions of the SDRAMs such as LPDDR2, etc.), RAMBUS DRAM (RDRAM), static RAM (SRAM), etc. One or more memory devices are coupled onto a circuit board to form memory modules such as single inline memory modules (SIMMs), dual inline memory modules (DIMMs), etc. Alternatively, the devices are mounted with a SoC or an integrated circuit in a chip-on-chip configuration, a package-on-package configuration, or a multi-chip module configuration.

(204) The peripherals **5304** include any desired circuitry, depending on the type of system **5300**. For example, in one embodiment, peripherals **5304** includes devices for various types of wireless communication, such as Wi-Fi, Bluetooth, cellular, global positioning system, etc. In some embodiments, the peripherals **5304** also include additional storage, including RAM storage, solid state storage, or disk storage. The peripherals **5304** include user interface devices such as a display screen, including touch display screens or multitouch display screens, keyboard or other input

devices, microphones, speakers, etc.

(205) As illustrated, system **5300** is shown to have application in a wide range of areas. For example, system **5300** may be utilized as part of the chips, circuitry, components, etc., of a desktop computer **5310**, laptop computer **5320**, tablet computer **5330**, cellular or mobile phone **5340**, or television **5350** (or set-top box coupled to a television). Also illustrated is a smartwatch and health monitoring device **5360**. In some embodiments, smartwatch may include a variety of general-purpose computing related functions. For example, smartwatch may provide access to email, cellphone service, a user calendar, and so on. In various embodiments, a health monitoring device may be a dedicated medical device or otherwise include dedicated health related functionality. For example, a health monitoring device may monitor a user's vital signs, track proximity of a user to other users for the purpose of epidemiological social distancing, contact tracing, provide communication to an emergency service in the event of a health crisis, and so on. In various embodiments, the above-mentioned smartwatch may or may not include some or any health monitoring related functions. Other wearable devices are contemplated as well, such as devices worn around the neck, devices that are implantable in the human body, glasses designed to provide an augmented and/or virtual reality experience, and so on.

(206) System **5300** may further be used as part of a cloud-based service(s) **5370**. For example, the previously mentioned devices, and/or other devices, may access computing resources in the cloud (i.e., remotely located hardware and/or software resources). Still further, system **5300** may be utilized in one or more devices of a home **5380** other than those previously mentioned. For example, appliances within the home may monitor and detect conditions that warrant attention. For example, various devices within the home (e.g., a refrigerator, a cooling system, etc.) may monitor the status of the device and provide an alert to the homeowner (or, for example, a repair facility) should a particular event be detected. Alternatively, a thermostat may monitor the temperature in the home and may automate adjustments to a heating/cooling system based on a history of responses to various conditions by the homeowner. Also illustrated in FIG. 55 is the application of system **5300** to various modes of transportation **5390**. For example, system **5300** may be used in the control and/or entertainment systems of aircraft, trains, buses, cars for hire, private automobiles, waterborne vessels from private boats to cruise liners, scooters (for rent or owned), and so on. In various cases, system **5300** may be used to provide automated guidance (e.g., self-driving vehicles), general systems control, and otherwise. These any many other embodiments are possible and are contemplated. It is noted that the devices and applications illustrated in FIG. 55 are illustrative only and are not intended to be limiting. Other devices are possible and are contemplated.

(207) The present disclosure includes references to “an “embodiment” or groups of “embodiments” (e.g., “some embodiments” or “various embodiments”). Embodiments are different implementations or instances of the disclosed concepts. References to “an embodiment,” “one embodiment,” “a particular embodiment,” and the like do not necessarily refer to the same embodiment. A large number of possible embodiments are contemplated, including those specifically disclosed, as well as modifications or alternatives that fall within the spirit or scope of the disclosure.

(208) This disclosure may discuss potential advantages that may arise from the disclosed embodiments. Not all implementations of these embodiments will necessarily manifest any or all of the potential advantages. Whether an advantage is realized for a particular implementation depends on many factors, some of which are outside the scope of this disclosure. In fact, there are a number of reasons why an implementation that falls within the scope of the claims might not exhibit some or all of any disclosed advantages. For example, a particular implementation might include other circuitry outside the scope of the disclosure that, in conjunction with one of the disclosed embodiments, negates or diminishes one or more the disclosed advantages. Furthermore, suboptimal design execution of a particular implementation (e.g., implementation techniques or

tools) could also negate or diminish disclosed advantages. Even assuming a skilled implementation, realization of advantages may still depend upon other factors such as the environmental circumstances in which the implementation is deployed. For example, inputs supplied to a particular implementation may prevent one or more problems addressed in this disclosure from arising on a particular occasion, with the result that the benefit of its solution may not be realized. Given the existence of possible factors external to this disclosure, it is expressly intended that any potential advantages described herein are not to be construed as claim limitations that must be met to demonstrate infringement. Rather, identification of such potential advantages is intended to illustrate the type(s) of improvement available to designers having the benefit of this disclosure. That such advantages are described permissively (e.g., stating that a particular advantage “may arise”) is not intended to convey doubt about whether such advantages can in fact be realized, but rather to recognize the technical reality that realization of such advantages often depends on additional factors.

(209) Unless stated otherwise, embodiments are non-limiting. That is, the disclosed embodiments are not intended to limit the scope of claims that are drafted based on this disclosure, even where only a single example is described with respect to a particular feature. The disclosed embodiments are intended to be illustrative rather than restrictive, absent any statements in the disclosure to the contrary. The application is thus intended to permit claims covering disclosed embodiments, as well as such alternatives, modifications, and equivalents that would be apparent to a person skilled in the art having the benefit of this disclosure.

(210) For example, features in this application may be combined in any suitable manner. Accordingly, new claims may be formulated during prosecution of this application (or an application claiming priority thereto) to any such combination of features. In particular, with reference to the appended claims, features from dependent claims may be combined with those of other dependent claims where appropriate, including claims that depend from other independent claims. Similarly, features from respective independent claims may be combined where appropriate.

(211) Accordingly, while the appended dependent claims may be drafted such that each depends on a single other claim, additional dependencies are also contemplated. Any combinations of features in the dependent that are consistent with this disclosure are contemplated and may be claimed in this or another application. In short, combinations are not limited to those specifically enumerated in the appended claims.

(212) Where appropriate, it is also contemplated that claims drafted in one format or statutory type (e.g., apparatus) are intended to support corresponding claims of another format or statutory type (e.g., method).

(213) Because this disclosure is a legal document, various terms and phrases may be subject to administrative and judicial interpretation. Public notice is hereby given that the following paragraphs, as well as definitions provided throughout the disclosure, are to be used in determining how to interpret claims that are drafted based on this disclosure.

(214) References to a singular form of an item (i.e., a noun or noun phrase preceded by “a,” “an,” or “the”) are, unless context clearly dictates otherwise, intended to mean “one or more.” Reference to “an item” in a claim thus does not, without accompanying context, preclude additional instances of the item. A “plurality” of items refers to a set of two or more of the items.

(215) The word “may” is used herein in a permissive sense (i.e., having the potential to, being able to) and not in a mandatory sense (i.e., must).

(216) The terms “comprising” and “including,” and forms thereof, are open-ended and mean “including, but not limited to.”

(217) When the term “or” is used in this disclosure with respect to a list of options, it will generally be understood to be used in the inclusive sense unless the context provides otherwise. Thus, a recitation of “x or y” is equivalent to “x or y, or both,” and thus covers 1) x but not y, 2) y but not x,

and 3) both x and y. On the other hand, a phrase such as “either x or y, but not both” makes clear that “or” is being used in the exclusive sense.

(218) A recitation of “w, x, y, or z, or any combination thereof” or “at least one of . . . w, x, y, and z” is intended to cover all possibilities involving a single element up to the total number of elements in the set. For example, given the set [w, x, y, z], these phrasings cover any single element of the set (e.g., w but not x, y, or z), any two elements (e.g., w and x, but not y or z), any three elements (e.g., w, x, and y, but not z), and all four elements. The phrase “at least one of . . . w, x, y, and z” thus refers to at least one element of the set [w, x, y, z], thereby covering all possible combinations in this list of elements. This phrase is not to be interpreted to require that there is at least one instance of w, at least one instance of x, at least one instance of y, and at least one instance of z.

(219) Various “labels” may precede nouns or noun phrases in this disclosure. Unless context provides otherwise, different labels used for a feature (e.g., “first circuit,” “second circuit,” “particular circuit,” “given circuit,” etc.) refer to different instances of the feature. Additionally, the labels “first,” “second,” and “third” when applied to a feature do not imply any type of ordering (e.g., spatial, temporal, logical, etc.), unless stated otherwise.

(220) The phrase “based on” is used to describe one or more factors that affect a determination. This term does not foreclose the possibility that additional factors may affect the determination. That is, a determination may be solely based on specified factors or based on the specified factors as well as other, unspecified factors. Consider the phrase “determine A based on B.” This phrase specifies that B is a factor that is used to determine A or that affects the determination of A. This phrase does not foreclose that the determination of A may also be based on some other factor, such as C. This phrase is also intended to cover an embodiment in which A is determined based solely on B. As used herein, the phrase “based on” is synonymous with the phrase “based at least in part on.”

(221) The phrases “in response to” and “responsive to” describe one or more factors that trigger an effect. This phrase does not foreclose the possibility that additional factors may affect or otherwise trigger the effect, either jointly with the specified factors or independent from the specified factors. That is, an effect may be solely in response to those factors, or may be in response to the specified factors as well as other, unspecified factors. Consider the phrase “perform A in response to B.” This phrase specifies that B is a factor that triggers the performance of A, or that triggers a particular result for A. This phrase does not foreclose that performing A may also be in response to some other factor, such as C. This phrase also does not foreclose that performing A may be jointly in response to B and C. This phrase is also intended to cover an embodiment in which A is performed solely in response to B. As used herein, the phrase “responsive to” is synonymous with the phrase “responsive at least in part to.” Similarly, the phrase “in response to” is synonymous with the phrase “at least in part in response to.”

(222) Within this disclosure, different entities (which may variously be referred to as “units,” “circuits,” other components, etc.) may be described or claimed as “configured” to perform one or more tasks or operations. This formulation—[entity] configured to [perform one or more tasks]—is used herein to refer to structure (i.e., something physical). More specifically, this formulation is used to indicate that this structure is arranged to perform the one or more tasks during operation. A structure can be said to be “configured to” perform some task even if the structure is not currently being operated. Thus, an entity described or recited as being “configured to” perform some task refers to something physical, such as a device, circuit, a system having a processor unit and a memory storing program instructions executable to implement the task, etc. This phrase is not used herein to refer to something intangible.

(223) In some cases, various units/circuits/components may be described herein as performing a set of task or operations. It is understood that those entities are “configured to” perform those tasks/operations, even if not specifically noted.

(224) The term “configured to” is not intended to mean “configurable to.” An unprogrammed FPGA, for example, would not be considered to be “configured to” perform a particular function. This unprogrammed FPGA may be “configurable to” perform that function, however. After appropriate programming, the FPGA may then be said to be “configured to” perform the particular function.

(225) For purposes of United States patent applications based on this disclosure, reciting in a claim that a structure is “configured to” perform one or more tasks is expressly intended not to invoke 35 U.S.C. § 112(f) for that claim element. Should Applicant wish to invoke Section 112(f) during prosecution of a United States patent application based on this disclosure, it will recite claim elements using the “means for” [performing a function] construct.

(226) Different “circuits” may be described in this disclosure. These circuits or “circuitry” constitute hardware that includes various types of circuit elements, such as combinatorial logic, clocked storage devices (e.g., flip-flops, registers, latches, etc.), finite state machines, memory (e.g., random-access memory, embedded dynamic random-access memory), programmable logic arrays, and so on. Circuitry may be custom designed, or taken from standard libraries. In various implementations, circuitry can, as appropriate, include digital components, analog components, or a combination of both. Certain types of circuits may be commonly referred to as “units” (e.g., a decode unit, an arithmetic logic unit (ALU), functional unit, memory management unit (MMU), etc.). Such units also refer to circuits or circuitry.

(227) The disclosed circuits/units/components and other elements illustrated in the drawings and described herein thus include hardware elements such as those described in the preceding paragraph. In many instances, the internal arrangement of hardware elements within a particular circuit may be specified by describing the function of that circuit. For example, a particular “decode unit” may be described as performing the function of “processing an opcode of an instruction and routing that instruction to one or more of a plurality of functional units,” which means that the decode unit is “configured to” perform this function. This specification of function is sufficient, to those skilled in the computer arts, to connote a set of possible structures for the circuit.

(228) In various embodiments, as discussed in the preceding paragraph, circuits, units, and other elements defined by the functions or operations that they are configured to implement, The arrangement and such circuits/units/components with respect to each other and the manner in which they interact form a microarchitectural definition of the hardware that is ultimately manufactured in an integrated circuit or programmed into an FPGA to form a physical implementation of the microarchitectural definition. Thus, the microarchitectural definition is recognized by those of skill in the art as structure from which many physical implementations may be derived, all of which fall into the broader structure described by the microarchitectural definition. That is, a skilled artisan presented with the microarchitectural definition supplied in accordance with this disclosure may, without undue experimentation and with the application of ordinary skill, implement the structure by coding the description of the circuits/units/components in a hardware description language (HDL) such as Verilog or VHDL. The HDL description is often expressed in a fashion that may appear to be functional. But to those of skill in the art in this field, this HDL description is the manner that is used transform the structure of a circuit, unit, or component to the next level of implementational detail. Such an HDL description may take the form of behavioral code (which is typically not synthesizable), register transfer language (RTL) code (which, in contrast to behavioral code, is typically synthesizable), or structural code (e.g., a netlist specifying logic gates and their connectivity). The HDL description may subsequently be synthesized against a library of cells designed for a given integrated circuit fabrication technology, and may be modified for timing, power, and other reasons to result in a final design database that is transmitted to a foundry to generate masks and ultimately produce the integrated circuit. Some hardware circuits or portions thereof may also be custom-designed in a schematic editor and

captured into the integrated circuit design along with synthesized circuitry. The integrated circuits may include transistors and other circuit elements (e.g., passive elements such as capacitors, resistors, inductors, etc.) and interconnect between the transistors and circuit elements. Some embodiments may implement multiple integrated circuits coupled together to implement the hardware circuits, and/or discrete elements may be used in some embodiments. Alternatively, the HDL design may be synthesized to a programmable logic array such as a field programmable gate array (FPGA) and may be implemented in the FPGA. This decoupling between the design of a group of circuits and the subsequent low-level implementation of these circuits commonly results in the scenario in which the circuit or logic designer never specifies a particular set of structures for the low-level implementation beyond a description of what the circuit is configured to do, as this process is performed at a different stage of the circuit implementation process.

(229) The fact that many different low-level combinations of circuit elements may be used to implement the same specification of a circuit results in a large number of equivalent structures for that circuit. As noted, these low-level circuit implementations may vary according to changes in the fabrication technology, the foundry selected to manufacture the integrated circuit, the library of cells provided for a particular project, etc. In many cases, the choices made by different design tools or methodologies to produce these different implementations may be arbitrary.

(230) Moreover, it is common for a single implementation of a particular functional specification of a circuit to include, for a given embodiment, a large number of devices (e.g., millions of transistors). Accordingly, the sheer volume of this information makes it impractical to provide a full recitation of the low-level structure used to implement a single embodiment, let alone the vast array of equivalent possible implementations. For this reason, the present disclosure describes structure of circuits using the functional shorthand commonly employed in the industry.

Claims

1. A memory apparatus, comprising: a plurality of bit cells, wherein the bit cells include a first set of first transistors formed in a first transistor region and a second set of second transistors formed in a second transistor region, the second transistor region being positioned below the first transistor region in a vertical dimension perpendicular to the plurality of bit cells, wherein the plurality of bit cells are divided into at least a first array of bit cells and a second array of bit cells; a first metal layer located below the plurality of bit cells in the vertical dimension, wherein the first metal layer includes first routing coupled to bitline outputs of the first array of bit cells; a second metal layer located above the plurality of bit cells in the vertical dimension, wherein the second metal layer includes second routing coupled to bitline outputs of the second array of bit cells; a first column input/output logic cell coupled to the bitline outputs of the first array of bit cells by the first routing; and a second column input/output logic cell coupled to the bitline outputs of the second array of bit cells by the second routing; wherein the first array of bit cells are positioned further away from the first column input/output logic cell and the second column input/output logic cell than the second array of bit cells.
2. The memory apparatus of claim 1, wherein outputs of the first column input/output logic cell are coupled to outputs of the second column input/output logic cell.
3. The memory apparatus of claim 1, wherein the bitline outputs of the first array of bit cells include outputs from a pair of complementary bitlines coupled to the first array of bit cells.
4. The memory apparatus of claim 1, wherein the bitline outputs of the second array of bit cells include outputs from a pair of complementary bitlines coupled to the second array of bit cells.
5. The memory apparatus of claim 1, wherein the bitline outputs of the first array of bit cells and the bitline outputs of the second array of bit cells are outputs of at least some first transistors.
6. The memory apparatus of claim 1, wherein the bitline outputs of the first array of bit cells are coupled to a pair of bitlines positioned in the second metal layer, the memory apparatus further

comprising: a first pair of connections between the bitlines in the second metal layer and the first routing in the first metal layer near the first array of bit cells.

7. The memory apparatus of claim 6, wherein the first pair of connections are positioned in a pair of first dummy cells near a boundary between the first array of bit cells and the second array of bit cells.

8. The memory apparatus of claim 1, further comprising: a second pair of connections between the second routing in the second metal layer and inputs of the second column input/output logic cell near the second column input/output logic cell, the inputs of the second column input/output logic cell being positioned in the second transistor region.

9. The memory apparatus of claim 8, wherein the second pair of connections are positioned in a second dummy cell near a boundary between the first column input/output logic cell and the second column input/output logic cell.

10. The memory apparatus of claim 1, wherein the first array of bit cells is adjacent to the second array of bit cells in a horizontal dimension perpendicular to the vertical dimension, and wherein the first column input/output logic cell is adjacent to the second column input/output logic cell in the horizontal dimension.

11. The memory apparatus of claim 1, wherein the first column input/output logic cell includes: a third set of first transistors formed in the first transistor region; and a fourth set of second transistors formed in the second transistor region; wherein inputs of the first transistors and inputs of the second transistors are merged and coupled to the bitline outputs of the first array of bit cells by the first routing.

12. The memory apparatus of claim 11, wherein gates of at least two of the first transistors are coupled by routing in the second metal layer.

13. The memory apparatus of claim 11, wherein the third set of first transistors includes two first transistors, and wherein the fourth set of second transistors includes five second transistors.

14. The memory apparatus of claim 1, wherein at least one of the bit cells includes four first transistors and two second transistors, the bit cell having: a first pass gate formed by a first one of the first transistors; a second pass gate formed by a second one of the first transistors; a first inverter formed by a third one of the first transistors and a first one of the second transistors; and a second inverter formed by a fourth one of the first transistors and a second one of the second transistors; wherein an input of the first inverter is cross-coupled with an output of the second inverter, and wherein an input of the second inverter is cross-coupled with an output of the first inverter.

15. An input/output logic cell for a memory apparatus, comprising: a pair of first transistors formed in a first transistor region, wherein the first transistors include first input source/drain regions, first output source/drain regions, and first gates; a set of five second transistors formed in a second transistor region, wherein the second transistors include second input source/drain regions, second output source/drain regions, and second gates, and wherein the second transistor region is positioned below the first transistor region in a vertical dimension; a first metal layer located below the second transistor region in the vertical dimension, wherein the first metal layer includes first routing coupled to bitline outputs of a first array of bit cells; and a second metal layer located above the first transistor region in the vertical dimension; wherein the first input source/drain regions of the first transistors and the second input source/drain regions of the second transistors are merged, and wherein the merged source/drain regions are coupled to the bitline outputs of the first array of bit cells by the first routing; wherein the gates of the pair of first transistors are coupled by routing in the second metal layer, and wherein the gates of at least two of the second transistors are coupled by routing in the first metal layer.

16. The input/output logic cell of claim 15, further comprising a second input/output logic cell for the memory apparatus, the second input/output logic cell comprising: a second pair of first transistors formed in the first transistor region; and a second set of five second transistors formed in

the second transistor region.

17. The input/output logic cell of claim 16, wherein the second metal layer includes second routing coupled to the bitline outputs of a second array of bit cells, and wherein the merged source/drain regions in the second input/output logic cell are coupled to the bitline outputs of the second array of bit cells by the second routing.

18. A memory apparatus, comprising: a plurality of bit cells, wherein the bit cells include a first set of first transistors formed in a first transistor region and a second set of second transistors formed in a second transistor region, the second transistor region being positioned below the first transistor region in a vertical dimension perpendicular to the plurality of bit cells, wherein the plurality of bit cells are divided into at least a first array of bit cells and a second array of bit cells; a first metal layer located below the plurality of bit cells in the vertical dimension, wherein the first metal layer includes first routing; a second metal layer located above the plurality of bit cells in the vertical dimension, wherein the second metal layer includes second routing coupled to bitline outputs of the second array of bit cells; a pair of bitlines positioned in the second metal layer, the pair of bitlines being coupled to the bitline outputs of the first array of bit cells; and a pair of connections between the bitlines in the second metal layer and the first routing in the first metal layer near the first array of bit cells.

19. The memory apparatus of claim 18, wherein the pair of connections are positioned in a pair of dummy cells near a boundary between the first array of bit cells and the second array of bit cells.

20. The memory apparatus of claim 19, wherein the pair of connections include trench metals in the pair of dummy cells.
